假大象与泡泡

公开数据、Bell/GHZ 与读数规则的审计手记（书稿汇编）

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1 封面图

正式设计稿（印刷 / 发布用）:

图上文案（与内文「工作副题」可并存：图偏量子主题，内文偏审计手记）

位置	英文	中文
主书名	THE ABSENT ELEPHANT AND THE BUBBLES	假大象与泡泡
副题	QUANTUM ENTANGLEMENT: BELL’S INEQUALITY & GHZ STATES	量子纠缠：贝尔不等式与 GHZ 态

> **Absent 与 Fake**
> 画面英文用 Absent（缺席的大象），序言与目录里常用「假大象」比喻——二者同一意象：**谁也没摸全的那头象。**对内文统一用词无冲突时，以外宣、封面英文 Absent 为准即可。

资源路径：`book/image/cover.jpg`（仓库内相对本文件为 `image/cover.jpg`）

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2 纯文字版式（无图预览、或扉页）

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3 假大象与泡泡

3.0.1 公开数据、Bell/GHZ 与读数规则的审计手记

The Absent Elephant and the Bubbles（与封面主英文对齐时）
The Fake Elephant and the Bubble（与序言「假大象」对齐时）

王 辉 · Hui Wang

2026

THE ABSENT ELEPHANT
AND THE BUBBLES
假大象与泡泡
QUANTUM ENTANGLEMENT: BELL'S INEQUALITY & GHZ STATES
量子纠缠：贝尔不等式与GHZ态
Bell ／ GHZ
贝尔

图 1: 《假大象与泡泡》封面

腰封一句（内文 pitch）

用球面泡泡的直觉搭一座假大象，再用公开记录问：换一把统计的尺子，故事会不会变。

3.1 书脊建议

假大象与泡泡
王辉 · 2026

3.2 封底文案（可选用）

本书从「光像不断胀大的泡泡、探测器只摸到一点」这一直觉出发，把同一套图景写进**格子上的可运行模拟**；继而用**公开实验数据**说明：**同一批记录，只改读数与后处理，Bell 型指标可以从约 1.1 跳到约 2.3。**全书把**模型内数值**与**对记录、协议、分母的审计**分开写清，并诚实标出**尚未钉死**的边界——不是盖棺定论的教科书，而是一份**可对照仓库复现**的手记。

相关仓库：`chain-explosion-model`（正文稿见 `book/` 目录）

---

4 Cover (English mirror)

4.1 Cover image

Final design (print / release):

On-image copy (can coexist with the in-book "audit notes" subtitle: the art leans quantum; the interior leans protocol audit)

Placement	English	Chinese
Main title	THE ABSENT ELEPHANT AND THE BUBBLES	假大象与泡泡
Subtitle	QUANTUM ENTANGLEMENT: BELL'S INEQUALITY & GHZ STATES	量子纠缠：贝尔不等式与 GHZ 态

> **Absent vs fake**
> The cover uses Absent (the elephant no one fully sees); the preface and TOC often say "fake elephant." Same image: **nobody has the whole elephant.** For outward-facing English, Absent on the cover is fine.

Asset path: `book/image/cover.jpg` (relative to this file: `image/cover.jpg`)

4.2 Text-only layout (no image preview, or half-title page)

图 2: Cover: The Fake Elephant and the Bubbles

5 The Fake Elephant and the Bubbles

5.0.1 Notes on auditing readout rules in public data and Bell/GHZ-style experiments

The Absent Elephant and the Bubbles (when aligned with cover English)
The Fake Elephant and the Bubble (when aligned with the preface metaphor)

Hui Wang

2026

Jacket one-liner (interior pitch)

Start from the intuition that light spreads like an expanding bubble while detectors sample a point; then ask the public record: if we change the statistical yardstick, does the story change?

5.1 Spine suggestion

```
The Fake Elephant and the Bubbles
Hui Wang · 2026
```

5.2 Back-cover copy (optional)

This book starts from the picture that **light behaves like an ever-growing bubble and a detector touches only a patch**, then implements the same image as **runnable lattice simulations**; it uses **public experimental data** to show that **on one fixed event table, changing readout and post-processing alone can move a Bell-type figure from about 1.1 to about 2.3**. **In-model numerics** are kept **separate** from **audits of records, protocols, and denominators**, and **unfinished** boundaries are labeled honestly —not a closed textbook, but a **repo-reproducible** lab notebook.

Repository: `chain-explosion-model` (manuscript under `book/`)

6 序言

6.1 假大象

假如光是一头大象。

几百年来，所有人都在摸这头大象，写下各自的测量结果。有人摸到了象腿，写出了经典力学；有人摸到了象耳，写出了波动方程；有人摸到了象鼻，写出了量子力学。每一套描述都是真实的，但没有人见过完整的大象。

我没有摸过真正的大象。我只是一个对未知好奇、对哲学感兴趣、对人类未来有一点危机感的普通人。我没有受过正规的物理学训练，我不低于任何人，也不高于任何人。我只是想做一件事：画一头假大象，让所有测量真大象的工具来测量我画的这头，看看哪些能通过，哪些不能。

通过的，说明我在那个维度画对了一些东西。没通过的，说明我的假大象在那里有缺陷——也可能问题不在「谁错了」，而在于：**记录与统计协议是否足够透明**：如何把连续响应映射成离散结果、事件如何配对、分母如何选取。

这些层面值得被认真审视；它与工程上可复现、预先固定规则的审计式阅读是同一类诚实，而不是对具体实验者的人身否定。

文稿导航（卷前与第一单元）
阅读顺序建议：`00-preface.md`（本序言）→ `01-appendix-three-phrasings.md`（三种口语、离散核，可选）→ 第 1–3 章正文（见 `02-contents.md` 路径）→ 第 4 章起遵 `00-unified-writing-scheme.md`；**Bell / GHZ 背景档案**见 `06-archive-bell.md`、`07-archive-ghz.md`。全书目录与章节文件见 `02-contents.md`。

零基础读者：章首不再设大块「导读框」，以免打断叙事；**专业词在正文里第一次需要处用一句话带过**即可。若你只想跟故事、不深究脚本，抓住各章主线仍读得下去；**想决定读到哪里为止**，可看**第 2、5、14 章**末尾的**分层路标**（见 `book/REVISION_PLAN_NARRATIVE_LAYERING.md` 方案说明）。

6.2 全书读法与边界（写在前面一次）

为避免**频繁打断**你的注意力，下列各条**在序言里写清一次**；**章内正文不再逐段复述**，除非当下句子离开这一条就会必然被误读。

1. **非法典**：本书不是「已写进教科书并由共同体盖棺定论」的定理汇编；推理、类比与数值示例**可错、可改**，以各章写明的定义、协议与数据来源为准。
2. **非本体论推销**：插图或口语里的「透明果冻」「连续背景」「介质」等，是为**眼睛与推演**服务的脚手架；**不**据此宣称「真空中存在机械以太」或替某种宇宙图景下判决书。
3. **非人身、非阴谋**：讨论的是记录、协议与统计对象是否透明，**不是**对具体实验者或机构的人身否定。
4. **与「反科学」脱钩**：声明边界，是为了**少误读**、把力气用在可复核的步骤上，而不是示弱或讨好。

涉及 Bell、GHZ 与公开数据的部分，我会把**模型内的数值实验**与**对公开记录与协议的审计式讨论**分开写清。公开测量文件对「连续物理图像」而言往往是**不完备的**；本书在承认这一点的前提下，仍讨论同一记录上**不同统计对象**会讲出怎样不同的故事。书中若出现 CHSH 等具体数值，**以各章对统计定义与数据来源的说明为准**，避免把序言里的比喻直接当成实验判决书。

我不回避失败：书里有章节与后记，诚实记录没通过与仍开放的部分。

这本书不挑战任何人。现有的物理学公式是人类几百年积累的智慧，我无意推翻它们。「推翻」这个词带着一种对抗的气息，而我内心没有对抗。我只是提供另一种语言——一种从球面波出发的语言——让有兴趣的人自己判断它是否有用。

我大概率是错的。这无法避免。

我的身体是宇宙元素的一次临时组合，我的思想是这次组合短暂产生的一点涟漪。人类一直在尝试认识自己，一直在修订昨日暂定的自画像——这本身就是我们存在的方式。我们是宇宙暂时借用的一种形态，我们和宇宙彼此包含。

所以这本书不是答案，是一次涟漪。它扩散出去，碰到什么，产生什么新的涟漪，我无法预知，也不需要预知。书里有未完成的地方，有开放的问题，有诚实的边界。我把这条口子挖开，后面的事交给后来者。

王　辉
2026 年 5 月 4 日

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7 Preface

7.1 A fake elephant

Imagine light is an elephant.

For centuries, everyone has been touching this elephant and recording their findings. Some felt the leg and wrote classical mechanics. Some felt the ear and wrote wave equations. Some felt the trunk and wrote quantum mechanics. Every description is real —but no one has seen the whole elephant.

I have never touched the real elephant. I am an ordinary person —curious about the unknown, drawn to philosophy, quietly worried about the human future. I have no formal physics training. I am not below anyone, and I am not above anyone. I simply want to do one thing: draw a fake elephant, and use every tool designed to measure the real one to test my drawing.

Where the measurements pass, I may have gotten something right. Where they fail, my drawing may have a flaw —or the issue may not be "who is wrong," but whether the **record and statistical protocol are transparent enough**: how continuous responses are mapped to discrete outcomes, how events are paired, and how denominators are chosen.

Those layers deserve scrutiny; they belong to the same kind of honesty as reproducible, preregistered audit-style reading, not a personal verdict on any experimenter.

Navigating the front matter (through Part I)
Suggested reading order: `00-preface.md` (this file) → `01-appendix-three-phrasings.md` (three phrasings of the model; optional) → Chapters 1–3 via the paths listed in `02-contents.md` → from Chapter 4 onward, follow `00-unified-writing-scheme.md`. **Bell / GHZ background archives**: `06-archive-bell.md`, `07-archive-ghz.md`. The full table of contents and chapter files: `02-contents.md`.

If you are not a physicist: there are **no** large chapter-opening "for general readers"boxes anymore — jargon is glossed **in-line** when it first matters. If you want to decide **how deep to read**, see the short **layer cues** at the ends of **Chapters 2, 5, and 14**, and `book/REVISION_PLAN_NARRATIVE_LAYERING.md`.

7.2 How to read this book —boundaries stated once

To avoid **constant footnotes** that pull you out of the narrative, the following is stated **once in this preface**; the main text **does not repeat** it paragraph by paragraph unless a sentence would otherwise be misread.

1. **Not a textbook of finished laws** —claims, analogies, and numbers can be wrong or revised; trust the definitions, protocols, and data sources spelled out in each chapter.
2. **Not selling an ontology** —visual aids ("clear jelly," "continuous backdrop," "medium") are scaffolding for intuition and calculation, **not** a claim that mechanical ether fills the vacuum or that a final picture of reality is settled.
3. **Not ad hominem** —scrutiny targets transparency of records and statistics, **not** personal verdicts on experimenters or labs.
4. **Not "anti-science" by posture** —boundaries exist to **reduce misreading** and keep effort on auditable steps.

Where Bell, GHZ, and public data appear, I separate **in-model numerical experiments** from **audit-style discussion of public records and protocols**. Published measurement files are often **incomplete as a full continuous picture**; still, the book asks what **different statistical objects** say when applied to the **same** record. Any headline CHSH-style numbers are **binding only to the definitions and data sources stated in each chapter**, not to metaphors in this preface.

I do not avoid failure: chapters and the afterword record what did not pass and what remains open.

This book challenges no one. The existing equations of physics represent centuries of human wisdom. I have no intention of overturning them. The word "overturn" carries a combative energy I do not feel. I only offer another language —a language that starts from spherical waves —and let interested readers judge for themselves whether it is useful.

I am most likely wrong. That cannot be avoided.

My body is a temporary arrangement of the universe's elements. My thoughts are a small ripple produced by that arrangement, briefly. Humanity has always tried to understand itself, always revised yesterday's working pictures —and that is simply how we exist. We are a form the universe borrows for a while. We and the universe contain each other.

So this book is not an answer. It is a ripple. Where it spreads, what it touches, what new ripples it creates —I cannot know, and I do not need to know. There are unfinished parts here, open questions, honest boundaries. I dig this opening. What comes next belongs to those who follow.

Hui Wang
May 4, 2026

8 附录一　三种表述：泡泡、涟漪与链式爆炸（真空介质动力学版）

泡泡、涟漪、链式爆炸是**同一套直觉**的三种**口语昵称**，不是三套宇宙论；下面这张表把它们对齐到离散公式与代码习惯。

> **导读与契约** 下文中，您将看到同一套物理图像在不同语境下使用不同的名称。**这不是三套彼此矛盾的理论，而是描述同一套“真空介质动力学”的三个直觉入口。**您可以选择最容易在脑海中成像的表述，再对照本章的公式与代码核对。

在本书的讨论中，我们使用**介质 / 连续背景**等说法，来描写扰动如何**局域地、一步一步**传递；这与序言中的边界一致：**它们是推演与数值实现的脚手架**，不等于宣称「真空中存在机械以太」或某种已判决的宇宙本体论。光并非飞行的小球，而是一种**可在此脚手架下表述的**扰动传递；因为空间是三维的且各向同性，直觉上常画成球面波前。数值实现里用二维格点近似时，会伴随由　与耦合系数体现的**逐步损耗与重分配。**

8.1 一、三种表述的核心直觉重点

表述	直觉重点	最易误解之处	与底层代码的关系
泡泡	扰动在三维空间中向外胀大的球面波前；遇到物质界面会产生次级波前	误以为“泡泡”仅仅是肥皂泡那样的二维薄膜结构	相干复数格点 `U` 的一步传播，强度表征为能量密度 `|U|^2`
涟漪	介质受到扰动后一圈圈传开的过程；状态从一格传递至相邻格点	误以为这只是二维平面水波，与三维光场无关	二维格点上的局域场量、邻域耦合与衰减系数
链式爆炸	能量爆发后，扰动以局域因果方式向多邻格分配、级联并分叉	误以为这是一种剧烈的化学爆炸或破坏性事件	`propagate_double_slit` 内核中的多向分配系数 `A, S, B`

8.2 二、从直觉到代码：介质动力学的数学表达

为了在计算中体现“介质动力学”的理念，我们不对麦克斯韦方程组或薛定谔方程做教条化的应用，而是将其转化为可运行的离散更新规则。

8.2.1 1. 介质与动力学传导

在离散实现中，我们用格点上的实值场量 **E_t(x, y)** 近似上述图像（`t` 为步数）。当某格被激发时，扰动按邻域规则**分配**到相邻格点，而不是「整团能量弹道式平移」。

8.2.2 2. 球面传递与损耗

三维直觉是球面波前；**与** `chain_explosion_numba.propagate_double_slit` **对齐的接收式**在格点上为 **λ 乘以若干邻格贡献之和**（全部为加；被 `barrier` 挡住的邻格不计入对应项）：

```
E[t+1](x,y) =   × (
      A·E[t](x-1, y) + B·E[t](x+1, y)
    + S·E[t](x, y-1) + S·E[t](x, y+1)
    + (S/2)·E[t](x-1, y-1) + (S/2)·E[t](x-1, y+1)
    + (S/2)·E[t](x+1, y-1) + (S/2)·E[t](x+1, y+1)
)
```

- **λ**：每步全局衰减（LAM）。
- **A, B, S**：沿 +x、−x、侧向及对角（折半）的**分配权重**，不是相减关系。

相干一步（`ce_engine_v3_coherent.propagate_coherent`）用复数格点 **U**，主方向为 +x、对角步长 **d = √2**：

```
U[t+1](x,y) =   × (
      A · exp(i·k) · U[t](x-1, y)
    + S · exp(i·k) · U[t](x, y-1) + S · exp(i·k) · U[t](x, y+1)
    + (S/2) · exp(i·k·d) · U[t](x-1, y-1) + (S/2) · exp(i·k·d) · U[t](x-1, y+1)
)
```

强度取 **|U|²**。可运行对照：appendix_model_three_intuitions_demo.py（仓库根目录：`python book/appendix_model_three_intuitions_demo.py`）。

8.2.3 3. 读出与统计

当场量到达探测器时，仪器作为介质边界的一部分，会进行局部取样。我们在此不对全场做非局域的坍缩假设，而是记录连续响应或经过阈值判定的离散信号。

8.3 三、术语使用指南

- **对外主名（本书）**：**泡泡 / 泡泡模型**（球面波前、次级波前的图像）。
- **口语辅助**：描写**水波式一圈圈**或**格点间一步一步**传递时，多用**涟漪**。
- **历史与项目名**：强调**多向分配、级联、分叉**时，可用**链式爆炸**（与仓库名 `chain-explosion-model` 同源）；与「泡泡」「涟漪」**指同一套离散核**，不是另一套理论。

请按阅读习惯任选其一建立直觉，再以本章公式与代码核对。实质都是：**同一套局域离散更新（加可选相干相位），用不同画面说话。**

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9 Appendix I •Three phrasings: bubbles, ripples, and chain explosion (vacuum-medium dynamics)

bubble, ripple, chain explosion are three **nicknames** for the **same discrete kernel** —this appendix is a **crosswalk** to formulas and code habits, not a second cosmology.

> **Guide and compact** Below you will see the same physical picture named differently in different contexts. **These are not three contradictory theories; they are three intuitive entry points into the same "vacuum-medium dynamics" story.** Pick the image that sticks, then check it against the formulas and code in this chapter.

Throughout the book we use **medium / continuous backdrop** language to describe how disturbances **propagate locally, step by step**; this matches the preface: **they are scaffolding for reasoning and numerics**, not a claim that "mechanical ether fills the vacuum" or that some final ontology has been decided. Light is not a flying pellet; it is a disturbance that **can be expressed on this scaffold**; because space is three-dimensional and isotropic, the natural cartoon is often a spherical

wavefront. When the implementation uses a 2D lattice, **gradual loss and redistribution** show up through and the coupling coefficients.

9.1 1. Core intuition for each phrasing

Phrasing	Intuition	Easy mistake	Tie to the underlying code
Bubble	A spherical front expanding in 3D; material interfaces spawn secondary fronts	Thinking a "bubble" is only a 2D soap film	Coherent complex lattice `U`, one propagation step; intensity as energy density `|U|^2`
Ripple	The medium carries a disturbance outward in rings; state hands off neighbor to neighbor	Thinking this is only 2D water waves, unrelated to a 3D light field	Local fields on a 2D lattice, neighborhood coupling, decay
Chain explosion	After a burst, the disturbance allocates locally to multiple neighbors, cascades, and branches	Thinking this means a chemical blast or destructive event	Multidirectional split coefficients `A, S, B` inside the `propagate_double_slit` kernel

---

9.2 2. From intuition to code: math for medium dynamics

To embody "medium dynamics" in computation, we do **not** dogmatically paste Maxwell or Schrödinger onto the page; we turn the picture into **runnable discrete update rules**.

9.2.1 1. Medium and dynamical transfer

In the discrete build, a real field **E_t(x, y)** on the lattice approximates the cartoon (`t` is the step index). When a site fires, the disturbance **allocates** to neighbors under local rules —not as a **rigid ballistic relocation** of a lump of energy.

9.2.2 2. Spherical transfer and loss

The 3D intuition is a spherical front; the **receiver-style** update aligned with `chain_explosion_numba.propagate_doubl` multiplies **λ by a sum of neighbor contributions** (all additive; neighbors blocked by `barrier` drop out of their terms):

```
E[t+1](x,y) =   × (
      A·E[t](x-1, y) + B·E[t](x+1, y)
    + S·E[t](x, y-1) + S·E[t](x, y+1)
    + (S/2)·E[t](x-1, y-1) + (S/2)·E[t](x-1, y+1)
    + (S/2)·E[t](x+1, y-1) + (S/2)·E[t](x+1, y+1)
)
```

- **λ**: global damping per step (LAM).

- **A, B, S**: **allocation weights** along +x, −x, sideways, and diagonals (halved) —**not** a subtraction recipe.

The coherent step (`ce_engine_v3_coherent.propagate_coherent`) uses complex lattice **U**, main direction +x, diagonal step **d = √2**:

```
U[t+1](x,y) =   × (
      A · exp(i·k) · U[t](x-1, y)
    + S · exp(i·k) · U[t](x, y-1) + S · exp(i·k) · U[t](x, y+1)
    + (S/2) · exp(i·k·d) · U[t](x-1, y-1) + (S/2) · exp(i·k·d) · U[t](x-1, y+1)
)
```

Intensity is **|U|²**. Runnable cross-check: appendix_model_three_intuitions_demo.py (repo root: `python book/appendix_model_three_intuitions_demo.py`).

9.2.3 3. Readout and statistics

When the field meets a detector, the instrument acts as part of the medium boundary and samples locally. We do **not** assume a nonlocal collapse of the entire field; we record continuous responses or thresholded discrete signals.

9.3 3. How to use the vocabulary

- **Primary outward name (this book)**: **bubble / bubble model** (spherical fronts and secondary fronts).
- **Colloquial helper**: **ripple** when you want the **concentric-ring** or **step-to-step-on-the-lattice** feel.
- **History and repo name**: **chain explosion** when you stress **multiway allocation, cascade, branching** (same root as `chain-explosion-model`); it names the **same discrete kernel** as "bubble" and "ripple," not another theory.

Pick whichever image reads best, then reconcile with the formulas and code here. Substantively it is all: **the same local discrete update (plus optional coherent phase), spoken with different pictures.**

10 第一章•Chapter 1

泡泡
The bubble

10.1 1.1 一块石头落进水里

你见过石头落进平静水面的样子。

石头落下去的那一刻，水面上出现了一个点。然后，从这个点开始，一圈一圈的纹向外扩散。

现在我问你：那一圈圈向外走的纹，是什么？

不是水从中心跑到了岸边。水分子几乎没有移动，它们只是在原地上下晃动。真正在走的，是一种状态——一种「这里刚刚被扰动过」的状态——从一个点传给了邻居，邻居再传给邻居的邻居。

那个向外走的圆圈，不是一个东西在飞。它是一个消息在传。

10.2 1.2 现在想象光

我想邀请你用同样的方式想象光。

不是一颗小球从灯泡里射出来，沿直线飞到你眼睛里。

而是：光源发出一个扰动，这个扰动像水面的涟漪一样，向外无限扩展——不是圆圈，而是球面，因为空间是三维的。一个不断膨胀的球面泡泡，从光源向外扩散，碰到什么就在那里产生一个新的泡泡，新的泡泡再继续扩散。

这就是本书的模型。非常简单，只有一张图：

- 光源发出一个球面泡泡。
- 泡泡碰到物体，产生新的泡泡。
- 探测器在泡泡经过的时候，读取到泡泡上的一个点。

就这三句话。

10.3 1.3 那个「点」是什么

注意最后一句：探测器读取到的，是泡泡上的一个点。

不是整个泡泡。不是泡泡的全部信息。只是泡泡在那一刻经过探测器时，留下的一个局部印记。

就像你站在岸边，水面的涟漪经过你的脚时，你感受到的只是那一刻、那一点的水面高低——而整个水面上其他地方发生的事，你完全感觉不到。

探测器读到的那个点，在物理教科书里有一个名字，叫做「光子」。

本书不打算争论这个名字对不对。本书只是想说：如果光其实是一个球面泡泡，那么「光子」不是泡泡本身，而是泡泡被局部采样的结果。

一个读数，不等于整张图。

10.4 1.4 这本书要做的事

我不知道这个模型是不是对的。

没有人摸过真正的大象，但每个人都在描述大象。我想画一头假大象，让所有测量真大象的工具来测量我画的这头——哪些能通过，哪些不能，诚实记录下来。

第一章到这里就结束了。几乎没有公式和数字；只借用了教科书里常见的「光子」一词，方便和后面的「一个读数」对照。

只有一张图：一个泡泡，碰到东西，产生新的泡泡，探测器读取一个点。

10.5 1.5 这本书不是一本哲学书

上面这张图，是本书的起点。但我不是在写哲学。

在后面的章节里，我会把这张图搬进**电脑里的格子世界**：光还是那颗「泡泡」，但每一步传播都会像传球一样**漏掉一点**（后面第三章专门讲这个）；障碍物、缝、探测器，都会变成格子上**看得见、改得动**的规则。

再往后，我会用同一套格子程序去对照一些**著名的物理实验**（例如 Bell 实验、GHZ 实验——名字陌生没关系，第二章会从故事讲起）。我想看的很简单：**当我们换一把「怎么读数、怎么统计」的尺子，结论会不会跟着变。**

第二章里你会看到一个**在本书锁定的数据源与处理定义下**复现的例子：同一批公开实验记录，只改一步后处理，一个关键数字会从大约 **1.1** 跳到大约 **2.3**。它不是「证明全世界怎样」，而是在提醒：**读数和规则若不写清，故事会跟着变。**

具体统计定义、脚本与数据路径，以**后文对应章节**为准；序言与第一章只搭架子。

泡泡、涟漪、链式爆炸三种口语指向同一套离散核，见 `book/01-appendix-three-phrasings.md`。

书里如果出现「NIST」「Bootstrap」这类词，都只是**数据来源**和**算不确定性的一种方式**；不懂可以跳过名词，抓住主线——**假大象要经得起同一套、写清楚规则的检验。**

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11 Chapter 1 •The bubble

泡泡
The bubble

11.1 1.1 A stone drops into still water

You have seen a stone hit a calm surface.

The instant it lands, a dot appears. Then rings move outward from that dot.

Ask yourself: what **are** those rings?

Not water traveling from center to shore. Molecules barely drift; they mostly bob in place. What travels is a **state** — "this patch was just disturbed" —passed to a neighbor, then the neighbor's neighbor.

The moving circle is not an object in flight. It is **news** on the move.

11.2 1.2 Now imagine light

Try the same move for light.

Not a tiny ball shooting out of a bulb in a straight line to your eye.

Instead: the source kicks the medium; the kick spreads outward like ripples —not a circle but a **sphere**, because space is 3D. An **expanding spherical bubble** leaves the source, hits objects, spawns new bubbles, and those bubbles keep spreading.

That is the model of this book. Three sentences:

- The source launches a spherical bubble.
- A bubble hitting matter spawns new bubbles.
- When a bubble sweeps a detector, the detector reads **one point** on it.

That is all.

11.3 1.3 What that "point" is

Note the last line: the detector reads **a point on the bubble**.

Not the whole bubble. Not global information. Just a **local imprint** as the bubble passes.

Standing on the shore, you feel only the height at your feet when a ripple crosses —you feel nothing of the rest of the surface.

In textbooks that local readout is often called a "photon."

This book will not fight over the name. The claim is simpler: if light is a spherical bubble, then a "photon" is **not** the bubble —it is what you get **sampling the bubble locally**.

One readout is not the whole map.

11.4 1.4 What this book tries to do

I do not know whether the model is right.

No one has touched the whole elephant, yet everyone describes it. I want to draw a **fake elephant** and run every tool meant for the real one —log what passes, what fails, honestly.

Chapter 1 stops here: almost no formulas or numbers; it borrows the textbook word "photon" only to contrast with "one readout."

One picture: a bubble hits things, spawns bubbles, a detector reads a point.

11.5 1.5 This is not a philosophy book

That picture is the starting point —but the book is not doing philosophy.

Later chapters move the image into a **lattice world on a computer**: still the same "bubble," but each propagation step **leaks** a little (Chapter 3); slits, barriers, and detectors become **explicit, editable rules** on the grid.

Still later, the same lattice code meets **famous experiments** (Bell, GHZ —unfamiliar names are fine; Chapter 2 tells the story). The question is blunt: **when we swap the yardstick for readout and statistics, does the conclusion move?**

Chapter 2 gives a **reproducible example under this book's locked data source and processing definition**: on one public record, a single post-processing change moves a key figure from about **1.1** to about **2.3**. It is not "proof about the whole world"; it is a reminder: **if readout and rules stay vague, the story moves.**

Exact statistical definitions, scripts, and data paths belong in **later chapters**; preface and Chapter 1 only frame the scaffold.

The three nicknames bubble / ripple / chain explosion point to the **same discrete kernel** —see `book/01-appendix-three-phrasings.md`.

Words like "NIST" or "bootstrap" are just **data sources** and **a way to quantify uncertainty**; if the jargon annoys you, skip it and hold the spine —**the fake elephant still has to pass the same, clearly written rules.**

12 第二章•Chapter 2

为什么我开始怀疑
Why I started asking questions

12.1 2.0 先说说我是谁

我不是物理学家，也没有受过任何正规的物理训练。我是一个普通人，平时对数字和逻辑感兴趣，业余时间会读一些科普文章、跑一些代码。

有一天，我在读关于双缝实验的科普——就是那个「光穿过两条缝，后面出现条纹」的实验。教科书的解释是「波粒二象性」：光有时候像波，有时候像粒子。我读了很多遍，总觉得这个解释像是在回避问题，而不是在回答问题。

于是我开始自己想：有没有更简单的图像，能解释同样的现象？

第一章的那个泡泡模型，就是我想出来的答案。我觉得它很美，很直接。我开始用代码把它实现出来，测试它能不能解释各种光学现象。

结果它通过了很多测试。有一次我随口问聊天程序：「这个模型能不能解释量子纠缠?」——对方说，不能，常见理由是 Bell 一类实验。**这只是我个人追问的起点，不是拿 AI 当权威论据。**

我不太服气，决定自己去找公开数据、用代码试。

12.2 2.1 模型很美，但被判了死刑

第一章里的泡泡模型，对我个人而言是一幅能算、能画的图像。可我很快遇到一个朴素问题：**别人凭什么要相信它？世界上有没有什么现成的检验，能对准这种「一圈圈往外递消息」的图像开枪?**

有人提醒我：有这一类东西，一般叫作 **Bell 实验**——它把两处相隔很远的读数放在一起，看它们「步调」能紧到什么程度；论文里常用 **S** 这样的数来描述那种紧度。**Bell 实验**在物理学里已经名气很大；在通行的理论表述里，它常被用来讨论：如果世界真的是像涟漪那样一步一步局域传递的，那么那种同步性在 **S** 这类指标上**过不了某道栏杆**。许多实验报告说栏杆被跨过去了。于是**在那一套叙述里**，我的球面泡泡模型像是被判了死刑。

我接受了这个判决，暂时放下了这个问题。

12.3 2.2 但我回来了

一段时间之后，我开始好奇：论文里那个 **S**（同步分数），到底是怎么算出来的?

我找到了一组公开的实验数据——**NIST**（美国国家标准与技术研究院）在 2015 年公开的一批实验记录；这里把它当作「这批数从哪下载」的名字就行。任何人都可以下载。我就下载了，然后开始用代码分析。

我做了一件很简单的事：用两种方式处理同一批数据，看看那个「同步分数」**S** 会不会不一样。

结果让我停下来很久。

12.4 2.3 同一批数据，两个不同的数字

那个衡量「两处读数有多步调一致」的数字，在论文里常记作 **S**（你把它想成一个**同步分数**就行）。许多介绍会说：S 超过 **2**，就像跨过了一道栏杆，用来支持「量子纠缠」的说法。

我用第一种方式处理数据——把探测器的输出保留为连续的数值——得到：

S = 1.117

我用第二种方式处理同一批数据——把探测器的输出强制读成 +1 或 -1 两个值——得到：

S = 2.297

同一批数据。同一组实验记录。只改了一个处理步骤：把连续的数字砍成两个值。

S 从 1.1 跳到了 2.3。

那道栏杆在 **S = 2**。很多人把「跨过去」当作量子世界的证据。但我发现：**跨过去的那一段高度，有很大一块是后来「把连续读数砍成 +1/-1」这一步抬出来的**——不是原始记录自己变高了，是我们换了一把尺子。

你当然会问：这两个数是不是「放之四海而皆准」?**不是。**它们来自**一批具体的公开记录 + 一套写进代码的后处理**。换数据子集、换符合窗、换归一化，数字会变——本书后面要做的，正是把**尺子**一条条摊开；**任何 headline 数字都以该章写明的定义为准**，不要把本章当成中学实验室的操作指南。

12.5 2.4 这让我想到了什么

我回想起第一章的那张图：探测器读取的，是泡泡上的一个点。

把连续的数字砍成 +1/-1——这一步，是不是就是把「球面泡泡上的一个局部读数」强制读成「一个完整的粒子」?

如果是，那么 S 值的跳跃，也许不是因为「量子纠缠很神秘」，而是因为「我们在测量的时候，用了一把不合适的尺子」。

这个问题，我对着数据和规则**查了好几个月**。后面各章，就是我沿着这个问题往下挖的过程。我不会只给你看两个数字，我会把每一把尺子都摆在桌上，让你自己去比。

下一章不接着吵 Bell，而是回到更底层：**格子程序里，能量为什么会「漏」、书里说的「热」指什么**——那是后面所有「洗掉条纹、改掉统计」的共同地面。Bell 与 GHZ 的**历史、不等式到底排除什么、实验长什么样**，见档案 `06-archive-bell.md`、`07-archive-ghz.md`（可与本章交叉阅读）。

路标：若你只想跟完「泡泡直觉 + 我为什么去盯公开数据」这条线，读完本章可以停一停透口气。要继续看格子上的损耗与测量怎么写成规则，顺着读第 3 章；若你主要冲 **NIST 档案和 CHSH 的尺子**来，也允许从目录**直跳第 8 章起**。你可以在这里停一停。后面的路还很长，但不急。

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13 Chapter 2 •Why I started asking questions

为什么我开始怀疑
Why I started asking questions

13.1 2.0 Who I am, briefly

I am not a physicist and have no formal physics training. I am an ordinary person who likes numbers and logic and sometimes reads popular science and runs code.

One day I was reading about the double-slit experiment —light through two slits, stripes on a screen. Textbooks invoke “wave–particle duality”: light is wave-like sometimes, particle-like other times. I reread that many times; it felt like **dodging the question** rather than answering it.

So I asked whether a **simpler picture** could explain the same thing.

The bubble model in Chapter 1 was my answer. I liked how direct it was. I coded it and tested it against optical phenomena.

It passed a lot of checks. Once I casually asked a chatbot whether the model could explain quantum entanglement —the usual answer was no, citing Bell-type tests. **That was only my personal starting point, not an appeal to AI as authority.**

I was stubborn enough to **pull public data and try it in code**.

13.2 2.1 A beautiful model, “sentenced”

Chapter 1’s bubble picture was, for me, something I could draw and compute. A blunt question followed: **why should anyone trust it —is there an off-the-shelf test that takes aim at this “hand the message along, ring by ring” image?**

People pointed me to what are usually called **Bell tests**: you compare two distant readouts and ask how tightly their steps can line up; papers often summarize that tightness with a figure like **S**. Bell tests are famous; in the standard theoretical packaging, if the world propagates **locally, step by step** —like my spherical-bubble picture —then a quantity like **S** cannot cross a certain bar. Many experiments report crossing it. **In that narrative**, my model looks **dead on arrival**.

I accepted the verdict and set the question aside —for a while.

13.3 2.2 But I came back

Later I wondered: in papers, how exactly is **S** computed?

I found a public dataset —**NIST** (the U.S. National Institute of Standards and Technology) released experimental records in 2015; here, treat that name as **where the CSV came from**. Anyone can download them. I did, and started analyzing in code.

I did something simple: process **the same batch** two ways and see whether **S** moved.

The result made me pause for a long time.

13.4 2.3 One dataset, two headline numbers

The figure that measures "how in-step two readouts are" is often called **S** (think **synchronization score**). Popular accounts say: if **S** clears **2**, a bar is crossed —evidence for "quantum entanglement."

First pipeline —keep detector outputs **continuous**:

S = 1.117

Second pipeline —**force** outputs into **+1 or −1**:

S = 2.297

Same batch. Same experimental record. One changed step: **chopping continuous numbers into two buckets.**

S jumped from 1.1 to 2.3.

The bar sits at **S = 2**. Many treat crossing it as evidence of the quantum world. What I saw: **a chunk of the "height" over the bar came from the step that hard-bins continuous readouts into ±1** —the raw record did not magically rise; **we swapped yardsticks.**

You should ask: are these two numbers universal? **No.** They come from **one specific public record + one pipeline frozen in code.** Change the subset, coincidence window, normalization —numbers move. Later chapters **lay out each yardstick**; **any headline figure follows the definition in that chapter** —do not treat this as a high-school lab recipe.

13.5 2.4 What that suggested

I thought back to Chapter 1: the detector samples **a point on the bubble.**

Chopping continuous numbers into ±1 —is that **forcing a local sample on a spherical bubble** into **a whole "particle"** ?

If so, the jump in **S** might owe less to "mysterious entanglement" and more to **measuring with the wrong ruler.**

I spent **months** on the data and the rules. The following chapters are the dig. I will not flash only two numbers; I will put **every ruler on the table** so you can compare.

Next chapter does not keep arguing Bell; it goes under the floor: **why energy "leaks" on the lattice and what "heat" means here** —the shared ground for later "washing out fringes" and "moving statistics." Bell and GHZ **history, what inequalities rule out, what experiments look like**: archives `06-archive-bell.md`, `07-archive-ghz.md` (can be read alongside this chapter).

Layer cue: If you only want the thread "bubble intuition + why I stared at public data," you can **pause here**. To follow loss and measurement-as-rules on the grid, continue Chapter 3. If you came mainly for **NIST files and CHSH yardsticks**, you may **jump in at Chapter 8**. You can rest here —the road ahead is long, and there is no rush.

14　第三章•Chapter 3

介质、损耗与热
Medium, loss, and heat

14.1　3.1　理想图与真实模拟

主画面可以略写细节，**一进具体程序就要把「漏」接上线。**

最简单的双缝演示在程序 `scripts/ce/ce_00_double_slit_demo.py` 里。每走一步，场上的数会先乘一个**小于 1** 的系数，代码里叫 **LAMBDA**，书里常写成 **λ**，这里你可以先记成 **「每步留下百分之多少」**——典型大约 **0.85**，也就是**每步大约丢掉一成多**（不是精确物理换算，只是量级直觉）。

没传下去的那一份，在格子里不再出现——你可以想象成传接力时球掉了一块、被草皮吃掉了；**掉到哪、变成什么温度**，我们多数脚本**没有细画**，只诚实说：**这里停笔了。**

14.2　3.2　三种「漏法」(故事版)

格子程序里，**漏**常见有三种摆法（可以叠在一起）：

(1) 每一步都按比例变瘦——λ（LAMBDA)
像全班传纸条，每人抄完撕掉一小角再往下传。除非程序里另外写了**一直在发光的源**，否则**不会凭空变胖。**

(2) 在某些格子多吞一口——吸收
经过缝口、吸收片、探测器那块区域时，再乘一个 **(1−η)**：η 越大，吞得越狠。条纹变浅，常常是你在路上**加了一块会吃能量的海绵**——**吃多少写在参数里**，不是事后说一句「神秘退相干」。

(3) 有的格子天生难走——「涩格子」
在 `scripts/explore/explore_disordered_media_scan.py` 这类实验里，有的格点像**泥洼**，传导打折扣。路还是那条邻格传递的路，但**难走的地方多了，远处就更暗、更花。**

自己问一句就行：这张图有没有「每步变瘦」？有没有「某块特别能吃」？光是不是一直在补充？

14.3　3.3　「热」在本书里 = 什么意思

不是说我们已经算清了每一摄氏度的来龙去脉。

在本书里，**「热」多半是一句短话**：指**没留在格子「可传播数字」里的那些份额。**它们**不再进入下一步**——叙事上可以想成一个**很大的环境口袋**（专业书常叫**热浴 / bath**），**口袋里面具体装着什么**，我们**多数时候没画。**

还有一种「热」是**故意捣乱**：程序里给数字加一点随机抖动，看结论会不会翻船——这叫**压力测试**，**不是**宣称算出了真实温度场。

另有一条名字里带 **thermal** 的审计脚本(`scripts/explore/ripple_quantum_tests/ripple_thermal_triplet_preregiste`
它用**另一套连续方程**玩「有损波动 + 参数能不能被规矩地找回来」的游戏，和 `ce_*` 那种**爬楼梯式格子**不是同一个式子，但**态度一样：损耗写清楚，规则写清楚。**

14.4　3.4　哪些算「已经做实」，哪些算「留口子」

程序里已经一而再、再而三用到的：

- **漏**可以用变瘦、吃一口、涩格子等方式**写进代码**；

- 漏法会改变**亮暗、条纹清不清、传得远不远**——后面 Bell/CHSH、GHZ 模拟、涟漪玩具曲线，都默认**世界会吃能量**，只是每章盯的侧面不同。

我们故意不说死、留给以后研究的：

- 很多旋钮还是**没有和现实米、秒一一钉死**的「纯数字把手」；
- 环境口袋里到底该装哪种微观图像、格子极限能不能和某条连续方程**严丝合缝对上**——**书里没有装成已经做完的定理。**

世界会饿，传播会瘦——这不是模型丢脸，而是**后面谈测量、谈统计时，默认站着的地面。**下一部分（从第 4 章起，见 `02-contents.md`）会把同一套「会吃能量的格子」接进**双缝、吸收体、延迟选择与 Bell/CHSH 玩具**——条纹怎么弱、曲线怎么弯，都站在本章这块地上；统计尺子怎么换，则接续第二章那条线。本章把**「会漏」**接进程序，与第一章主画面形成**有意的分工**，而不是互相打架。

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15 Chapter 3 •Medium, loss, and heat

介质、损耗与热
Medium, loss, and heat

15.1 3.1 Ideal pictures vs honest simulation

The main cartoon can stay clean; **once you run code, plug in the leak.**

The minimal double-slit demo lives in `scripts/ce/ce_00_double_slit_demo.py`. Each step multiplies the field by a **number below 1**, called **LAMBDA** in code and **λ** in the book —shorthand: **"what fraction survives each step."** A typical value is about **0.85**, i.e. **you lose a bit more than ten percent per step** (order-of-magnitude intuition, not a calibrated physical conversion).

What does not propagate onward simply disappears from the lattice —like dropping part of the baton into the grass; **where it went and what temperature it became** is **not detailed** in most scripts —we **stop the story there**, honestly.

15.2 3.2 Three ways to "leak" (story version)

Lattice code usually implements **loss** in three combinable ways:

(1) Uniform thinning every step —λ (LAMBDA)
Like passing a note: each person tears off a corner before forwarding. Unless the code adds a **steady source**, nothing **grows from nothing**.

(2) Extra bites at certain sites —absorption
At slits, absorber sheets, or detector regions, multiply by **(1−η)**; larger **η** means a hungrier sponge. **Shallower fringes** often mean you **put a sponge on the path** —**how much is a parameter**, not a post-hoc "mysterious decoherence."

(3) Intrinsically rough sites — "sticky" cells
In scans like `scripts/explore/explore_disordered_media_scan.py`, some sites act like **mud** —coupling is discounted. The neighborhood rule is the same, but **more rough patches** make the far field **dimmer and messier**.

Sanity check: Does this picture have per-step thinning? Local hungry patches? A steady source?

15.3 3.3 What "heat" means in this book

Not that we have traced every Celsius degree.

Here **"heat" is usually shorthand** for **whatever share no longer lives in the lattice's "propagating numbers."** It **does not enter the next step** —narratively, think of a **huge environment pocket** (textbooks say **thermal bath**); **what exactly sits inside** is **mostly not drawn**.

Another "heat" is **deliberate noise**: jitter added to stress-test whether conclusions flip —**a robustness test**, **not** a claim about a real temperature field.

There is also an audit script with **thermal** in the name (`scripts/explore/ripple_quantum_tests/ripple_thermal_triplet_preregistered_audit.py`): it plays a **different continuous equation** game ("lossy waves + can parameters be recovered cleanly"). It is **not** the same formula as the **stair-step** `ce_*` **lattice**, but the **attitude matches: write loss, write rules.**

15.4 3.4 What is "done" vs what stays open

Already used again and again in code:

- **Leak** can be coded as thinning, local bites, sticky cells;
- leak changes **brightness, fringe clarity, reach** —later Bell/CHSH, GHZ sims, and ripple toy curves all assume **the world eats energy**, focusing on different sides each chapter.

Deliberately not closed:

- many knobs are still **pure numbers**, not pinned to real meters and seconds;
- what belongs inside the environment pocket, whether the lattice limit **locks cleanly** to a chosen continuum equation —**the book does not pretend those theorems are finished.**

The world gets hungry; propagation gets thin —not a shame for the model, but **the floor assumed when we later talk measurement and statistics.** From Chapter 4 on (see `02-contents.md`), the same **energy-eating lattice** meets **double slits, absorbers, delayed choice, and Bell/CHSH toys** —fringe weakening and curve bending stand on this ground; swapping statistical yardsticks continues Chapter 2's thread. This chapter wires **leakage** into the program, **on purpose alongside** Chapter 1's main picture —they **divide labor**, they do not fight.

16 档案一•Bell 实验与 Bell 不等式

人话一句：**Bell 不等式**像一道「本地小抄」能解释的关联统计**上限**；实验若反复超过这道上限，就要放弃**那一类**局域预先赋值故事。下文把 **EPR、Bell、Aspect** 以降的脉络和**实验骨架**写清，后文用到的 **CHSH** 是四人姓氏缩写的一种常用检验组合；**公式**在后文，**核心是比较界与观测。**

文稿性质：供本书读者对照的背景档案；**不是**对当代实验的完整综述，也不是对任何实验室的判决书。具体数值与协议以各章及原始论文为准。

16.1 A. 历史脉络：问题从哪来

1935，EPR
爱因斯坦、波多尔斯基、罗森提出一个思想实验：若量子力学完备，某些「纠缠」粒子对似乎允许**超距关联**；他们相信自然界底层应是**局域的**（一边发生的事不应瞬间决定另一边的结果），因而怀疑量子力学的描述是否完备。

玻尔与哥本哈根传统
强调测量与互补性，不把「同时赋予粒子确定位置与动量」当作合法图景。

隐变量思路
若存在**我们尚未看见的额外变量**（隐变量），粒子在测量前就可能带有「预定结果」，表观随机只是对隐变量的无知——这样世界仍可想象成**局域、经典式因果**的，只要这些隐变量按局域规则分布。

Bell 登场前的气氛
许多人直觉上希望：存在一种**局域、实在**的底层图像，能复现量子预言。Bell 要追问的是：**这到底行不行？能不能用数学钉死边界？**

16.2 B. Bell 做了什么：不等式是什么，不是什么

1964，John Bell
在一篇现在已成为经典的论文里，Bell 关心的不是「量子力学对不对」，而是：

> 若世界由**局域隐变量**按某种经典概率规则产生结果，且满足**局域性**（一边的测量设定不能瞬间改变另一边的隐变量分布），那么某些**可观测的相关量**必须满足一组**不等式**。

这就是 **Bell 型不等式**的家族。最常在实验里出现的一种变形叫 **CHSH 不等式**（四人名字 Clauser–Horne–Shimony–Holt）：用四个计数率组合成一个量 **S**，在一大类**局域隐变量模型**里必须满足 **|S| ≤ 2**（书中第二章用 **S** 讲故事时，指的就是这一类对象；具体定义依赖**如何把连续读数变成 ±1、事件如何配对、分母怎么取**——所以同一批原始记录可以算出不同的 **S**）。

Bell 不等式证明了什么？

- **证明了**：在明确写出的前提（局域性 + 隐变量产生结果的通常假设）下，**有一类数学结构必须成立**；量子力学对某些纠缠态的预言**会打破**这类结构。
- **没有证明**：「意识」「超光速信号」或某一种解释学；也没有单独「证明量子力学终极正确」——它是对**一类经典式局域模型**的**排除**（在实验与附加假设满足时）。

一句话：Bell 给「局域经典式因果 + 通常隐变量」画了一条**红线**；标准量子纠缠预言**踩过红线**。

16.3 C. 「实验」在测什么：思想骨架

典型 **Bell 测试**（以两光子偏振为例）：

1. **制备**：产生一对在某种意义上「关联」的光子（实验室宣称接近理论上的纠缠态）。
2. **分开**：光子 **A**、**B** 飞向两处，空间上拉开，保证类空分离（尽可能关**局域性漏洞**）。
3. **独立选择测量方向**：Alice 随机选一组分析角度，Bob 随机选另一组；角度在飞行中或临近测量前选定（与**测量设定独立性**、关闭**自由选择漏洞**有关）。
4. **记录**：每次符合得到计数；累积统计。
5. **构造 S 或类似量**：把计数组合成 Bell 不等式里出现的那类表达式。
6. **对比**：若 **S**（或等价量）稳定地**超过经典界**（CHSH 里常取 **2** 为讨论栏杆），在**标准诠释 + 标准漏洞关闭叙事**下，就**不利于**「局域隐变量按那种方式产生关联」的简单图像。

常见「漏洞」（读者在新闻里会见到）：

- **探测效率 / 探测漏洞**：探测器只抓到一部分事件，若子样本有偏，经典模型有时可「模仿」违反。

- **局域性漏洞**：若测量设定或结果仍能以低于光速的方式串通，不等式的前提被破坏。
- **记忆 / 记忆型隐变量**：若允许设备跨轮次「记仇」，讨论要换模型类。

2015 年前后，多个实验组宣称在**同时关断主要漏洞**的前提下观察到违反（不同平台：光子、金刚石色心、超导等）。学界对「哲学上是否已彻底说服所有人」仍有讨论，但**技术共识**大致是：在当代接受的假设包下，**局域隐变量按 Bell 那类方式组织**已极难自洽。

16.4 D. 已做过哪些类型的实验（档案式列举）

下列是**类型学**，不是完整文献表；年份与组名便于你按关键词检索。

类型	大致内容
早期光学	20 世纪 70–80 年代起，用光子偏振做 CHSH；Aspect 等（1982）缩小局域性漏洞关切。
离子阱	用囚禁离子做纠缠与 Bell 违反，探测效率高、态制备可控。
光子 + 高效率探测器	向关闭探测漏洞推进。
随机数 + 类空分离	用遥远随机源选测量基，强化「设定独立性」叙事。
** loophole-free / 全主要漏洞关闭 **	2015 年前后多个实验室报道（不同物理平台）。
表观违反经典界的大规模网络与场测	量子通信、设备无关协议等工程方向延伸。

与本书仓库的交点：第二章讨论的 **NIST 公开记录**属于「把同一批数据用不同后处理管道算 **S**」——这**不替代**上述实验物理学史，但**承接** Bell 测试对**统计对象与协议**的敏感性。

16.5 E. 与「泡泡 / 局域传递」模型的关系（本书立场）

本书第一章的泡泡图像，在**叙事上**强调：**扰动局域地、一步一步传递**；第二章又交代：在**通行的 Bell 叙述**里，这类**局域经典因果 + 通常隐变量**的打包，很难与**已报道的纠缠违反**兼容。

因此：Bell 实验在当代话语里，常被说成对「你这种直观」的**死刑判决**——本书第二章接受这句话的**情绪**，但把**数字怎么算**拉回可审计地面。

读者应带走的三点

1. **Bell 不等式**是**数学**：在给定前提下，局域隐变量必须满足某条界。
2. **实验**是在**尽力满足那些前提**的前提下，看自然界是否**踩界**。
3. **本书**关心：踩界叙事里，**尺子与分母**是否总已摊开——这与 Bell 的精神（把假设写清）**同向**，而不是反科学。

16.6 F. 推荐阅读入口（自行深入）

- Bell 原论文及现代教科书中的 **CHSH** 推导。
- 综述与科普：以「局域隐变量」「loophole-free」为关键词检索近年 **Review**。

- 本书后文：栅格 **Bell/CHSH 玩具**（`ce_bell*`）与公开数据审计章节——**模型内数值**与**对记录的审计**分开阅读。

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17 Archive I •Bell tests and Bell inequalities

In one line: a **Bell inequality** is an upper bound on certain correlations if outcomes were fixed by **local hidden instructions**; experiments that keep crossing it force you to drop **that** local story. Below: **EPR → Bell → Aspect-era** history, what labs actually measure, and **CHSH** as a common four-setting packaging —**formulas** follow, but the spine is **bound vs observation**.

Nature of this text: background for readers of this book; **not** a full survey of contemporary experiments and **not** a verdict on any lab. Numbers and protocols follow each chapter and the primary literature.

17.1 A. Historical thread: where the problem came from

1935, EPR
Einstein, Podolsky, and Rosen posed a thought experiment: if quantum mechanics is complete, some "entangled" pairs seem to allow **nonlocal correlations**; they believed nature's substrate should be **local** (what happens on one side should not instantly fix the other), so they questioned whether quantum mechanics is complete.

Bohr and the Copenhagen line
Stress measurement and complementarity; assigning simultaneous definite position and momentum to one particle is not treated as a valid picture.

Hidden-variable hope
If **extra variables we do not see** exist, outcomes might be **predetermined** before measurement; apparent randomness is ignorance of those variables —the world could still look **locally, classically causal** if the variables obey local rules.

Before Bell
Many hoped for a **local, realistic** substrate that reproduces quantum predictions. Bell asked: **can this work at all —can math draw a hard line?**

17.2 B. What Bell did: what the inequality is —and is not

1964, John Bell
In what became a classic paper, Bell was not asking "is quantum mechanics right?" but:

> If the world is produced by **local hidden variables** under ordinary classical probability rules, and **locality** holds (one side's measurement setting cannot instantly retune the other side's hidden-variable distribution), then certain **observable correlations** must obey **inequalities**.

That is the Bell family. A common experimental variant is **CHSH** (Clauser–Horne–Shimony–Holt): four rates combine into **S**; a broad class of **local hidden-variable models** requires **|S| ≤ 2** (when Chapter 2 uses **S**, it means this kind of object; the exact definition depends on **how continuous readouts become ±1, how events pair, how denominators are chosen** —so one raw table can yield different **S**).

What Bell inequalities prove

- **They prove**: Under stated premises (locality + ordinary hidden-variable outcome generation), **a mathematical structure must hold**; quantum predictions for some entangled states **break** that structure.

- **They do not prove**: "consciousness," "superluminal signals," or any one interpretation; nor do they alone "prove quantum mechanics forever"—they **rule out a class of classical local models** (when experiments and auxiliary assumptions hold).

One line: Bell drew a **red line** for "local classical causality + ordinary hidden variables"; standard entanglement predictions **step over** that line.

17.3 C. What a "Bell test" measures: conceptual skeleton

A typical **Bell test** (two photons, polarization):

1. **Prepare** a pair "correlated" in the lab's claimed sense (near an entangled state).

2. **Separate** photons **A** and **B** spatially for spacelike separation (closing the **locality loophole** as far as possible).

3. **Choose bases independently**—Alice picks an analysis angle, Bob another; angles are set in flight or just before measurement (ties to **measurement-setting independence** and the **free-choice loophole**).

4. **Record** coincidences per trial; accumulate statistics.

5. **Build S or an equivalent** from counts.

6. **Compare**—if **S** stably **beats the classical bar** (CHSH often uses **2** as the rhetorical fence), under **standard interpretation + standard loophole narrative**, that **disfavors** the simple picture of "local hidden variables producing correlations that way."

Common "loopholes" (you will see these in news):

- **Detection / fair-sampling**: if detectors miss many events, biased subsamples can let classical models mimic violations.

- **Locality**: if settings or outcomes can still coordinate subluminally, the inequality's premises fail.

- **Memory / memory-type hidden variables**: if devices "remember" across rounds, the model class changes.

Around 2015, several groups reported violations with **major loopholes closed** in one run (photons, NV centers, superconducting qubits, etc.). "Has everyone been philosophically persuaded?" is still debated; the **technical consensus** is roughly: under accepted auxiliary packages, **local hidden variables organized Bell's way** are extremely hard to sustain.

17.4 D. Types of experiments (taxonomy, not a full bibliography)

This is a **type list**, not exhaustive literature; years and group names help keyword search.

Type	Rough content
Early optics	Photons, CHSH from the 1970s–80s; Aspect et al. (1982) narrowed locality concerns.
Ion traps	High control, high detection efficiency for entanglement and Bell violations.
Photons + high-efficiency detectors	Push toward closing the detection loophole.
Random numbers + spacelike separation	Distant random sources for bases, strengthening "setting independence" narratives.
Loophole-free / all-major-loopholes-closed	Multiple labs ~2015 (different platforms).
Large network / field demos	Quantum communication, device-independent protocols, engineering extensions.

Intersection with this repo: Chapter 2's **NIST public record** is "**same data, different pipelines for S**"—it **does not replace** experimental history, but it **inherits** Bell tests' sensitivity to **statistical objects and protocols**.

17.5 E. Relation to the "bubble / local propagation" picture (this book)

Chapter 1's bubble story **narratively** stresses **local, step-by-step propagation**; Chapter 2 adds: in the **standard Bell narrative**, that package of **local classical causality + ordinary hidden variables** struggles with **reported entanglement violations**.

So: Bell tests are often called a **death sentence** for "your intuition"—this book accepts the **emotional weight** of that line, but pulls **how the number is computed** back onto auditable ground.

Take away three things

1. **Bell inequalities** are **math**: under given premises, local hidden variables must obey a bound.
2. **Experiments** ask whether nature **crosses the bound** while **trying to satisfy those premises**.
3. **This book** asks whether **yardsticks and denominators** are always laid bare—aligned with Bell's spirit (write assumptions clearly), **not** anti-science.

17.6 F. Pointers if you want to go deeper

- Bell's original paper and **CHSH** derivations in modern textbooks.
- Reviews and popular pieces: search **local hidden variables**, **loophole-free**.
- Later in this book: grid **Bell/CHSH toys** (`ce_bell*`) and public-data audit chapters—read **in-model numbers** and **record audits separately**.

18 档案二•GHZ 实验与 GHZ 态：为什么它「否定」我们这一类模型

人话一句：**GHZ** 是 Greenberger、Horne、Zeilinger 三人姓氏缩写——可以想成**三人版「谁的小抄对不上号」**：若坚持每人测量前就有固定答案且不能隔空商量，会推出**自相矛盾**；比两体 Bell 更「一刀见骨」，但真实实验仍有噪声与漏洞。后面 **in-silico** 章是**电脑里的协议游戏**，**不能**顶替实验室原始迹。

文稿性质：背景档案；把**标准量子力学框架里** GHZ 与**局域经典图像**的冲突说清楚。本书的泡泡/栅格模型属于**局域递推**一类直觉；读完本篇，你应能精确说出：**被否定的是哪一类陈述**，**不是**什么。

18.1 A. GHZ 是谁，解决了什么问题

1989，Greenberger、Horne、Zeilinger（及 Mermin 的清晰表述）
Bell 定理对**两体纠缠**给出**统计性**违反：要累积很多次测量，才能看到**平均上**超过 Bell 界。

GHZ 构造则展示：对**三体或更多体**的特定纠缠态（**GHZ 态**），在理想条件下，可以出现一个**逻辑上的尖锐矛盾**——

> 若坚持「每个粒子在测量前就有确定的、与远处设定无关的答案」（一种**非语境的、局域实在**式的赋值），
> 那么一组**本应同时成立**的等式，会**互撞车**（**0 = 1** 式的矛盾）。

这被称为 **GHZ 悖论**或 **GHZ 定理**：它把「局域隐变量按那种方式给结果」逼到**单轮论证**里就**站不住**，而不必等大数据统计（当然，真实实验仍有噪声与缺陷）。

18.2 B. GHZ 态（极简版）

三量子比特的 **GHZ 态**常写成：

```
|GHZ ⟩ = (|000⟩ + |111⟩) / √2
```

(基矢记号仅作示意；本书不在这里教量子计算。)

直觉：它不是「三个各自独立的硬币」；也不是「A 定了 B 就跟着定」那种经典关联能轻易穷尽的。它是**三体一起**才能正确描述的一种**纠缠**。

18.3 C.「证明了什么」——分三层说清楚

第一层（数学-逻辑层，理想 GHZ）
在**标准量子规则**下，对某些**互相对易的可观测量**的选择组合，预言一组**完美相关**；若强行给每个粒子预先指定 **+1 / −1** 型的「客观答案」（且答案不依赖远处**本轮**选测什么——一种常见的**局域隐变量**想象），这些答案**无法同时**满足所有关系。**因此：这一类「预先定好、局域携带」的赋值模型被排除。**

第二层（实验层）
真实实验室用离子、光子、超导比特等**逼近** GHZ；读出的是**有限保真度**的数据，要扣噪声、串扰、丢失。结论通常是：**观测与标准量子预言一致**，与「简单局域赋值表」不一致——但**每一句话都要带误差棒与协议**。

第三层（哲学层——本书不替你下判决书）
GHZ **没有**证明「意识创造实在」或某一种诠释；它**尖锐地打击**的是：**把多体纠缠当成三枚各自有预定答案、只靠经典关联串起来的硬币**——那种图像在理想论证里**直接崩**。

18.4 D. 实验大致怎么做（档案骨架）

1. **制备多体纠缠**：例如 trapped ions 用门操作把三离子推到接近 |GHZ⟩。
2. **选基测量**：每人（每离子）可选不同方向；记录 **+1 / −1** 或等价二元结果。

3. **检验预言的相关**：看联合概率与 GHZ 预言是否一致；或与 **Mermin 型不等式**对比（多体 Bell-type 界）。

4. **与经典界对照**：若设备与协议可信，违反**多体 Bell / Mermin** 界与违反**局域赋值模型**的叙事并行（细节见专业综述）。

与 Bell 两体案的区别
- Bell：**统计积累**见分晓。
- GHZ：**理想极限下**可在**逻辑链条**上见分晓；实验上仍要谈**有限数据与噪声**。

18.5 E. 已做过哪些方向（类型列举）

方向	说明
离子阱	多离子 GHZ、全纠缠门集，保真度与读出较可控。
光子	多光子纠缠，难在损耗与符合计数；有 GHZ 与类 GHZ 报道。
超导量子比特	中等规模芯片上的多体纠缠与量子优势演示路线。
与 Bell 测试结合	多体不等式、设备无关协议等。

本书仓库中 `scripts/explore/ghz_medium_v10` 等属于**模型内 in-silico 审计**，**不是**上述实验室原始迹的替代品——档案与代码要**分开读**。

18.6 F. 它为什么「否定我们的模型」——把话说到最清

本书模型（泡泡、格点递推、局域耦合）在**叙事上**承诺：

- 信息/扰动**局域地**从一格传到邻格；
- 「远处」的结果不应依赖一种**即时、非局域**的协调，除非你把那种协调**显式写进**新的场或新的全局变量（那就**不再是**原来那种「只靠局域泡泡碰泡泡」的最小图像）。

标准 GHZ 论证所打击的，正是：

- **给每个粒子预先分配确定的测量答案**；
- 且这些答案**不随远处本轮测量基的选择而「语境式」地整体改口**（否则你引入的是另一种理论，不是那张最简单的「局域藏宝图」）。

因此：

陈述	是否被 GHZ「否定」
「三个粒子各有一张**预先印好的经典答案卡**，且与远处当次选基无关」	**是**，理想 GHZ 论证下**很难保住**。
「任何**经典场**、任何**耗散**、任何**介质**只要写得够花就能自动逃掉」	**否**，不是一句话豁免；你要**具体写出**能否复现 GHZ 相关；本书第三章的损耗**没有**单独给这张豁免状。
「本书的**离散栅格代码**是否已经**复现**标准 GHZ 实验的全部协议与统计」	**否**，仓库自有边界；见各 **ghz**_* 脚本的说明与失败记录。

一句话给作者自己的模型：
若坚持**只有**局域递推、**没有**允许多体在测量事件上按量子规则非平凡关联，则**标准 GHZ 型纠缠所展示的那类多体相关**，与「最小局域经典赋值」**冲突**——这就是**「否定」的精确含义**：**否定那张过于简单的经典藏宝图**，而不是自动否定你**将来**是否可能写出**另一类**仍局域但更丰富的理论（那要另起炉灶并接受新实验的检验）。

18.7 G. 与第二章、第三章的衔接

- **第二章**：同一批数据，换**二值化尺子**，**S** 会跳——提醒 **Bell/CHSH 层**的协议敏感。
- **第三章**：介质会**吃能量**——但**吃能量本身**不自动解决 **GHZ 赋值矛盾**；那是**相关结构**问题，不只是**幅度衰减**问题。
- **本篇档案**：当你说「GHZ 否定了泡泡模型」时，请说成：**否定了「只靠局域递推 + 简单预先赋值」那一版最小图像**；本书后文若讨论 **in-silico GHZ**，那是在问**另一套**显式规则能不能、在多大程度上**逼近**公开叙事——**与本篇档案并行**，互不顶替。

18.8 H. 深入入口

- Mermin 对 GHZ 的「三硬币悖论」式科普表述。
- 多体 **Bell 不等式**、**Mermin–Klyshko** 界。
- 本书目录中 **第 11 章**（GHZ 与 in-silico 协议审计）及 `ghz_medium_v10` 索引。

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19 Archive II •GHZ tests and GHZ states: what exactly "rules out" models like ours

In one line: **GHZ** (Greenberger–Horne–Zeilinger) is a **three-body "answer-sheet contradiction"** — sharper than two-body Bell in logic, though real data still carry noise and loopholes. Later **in-silico** chapters are **protocol games on a computer**, **not** raw lab traces.

Nature of this text: background —clarifies the clash between GHZ in **standard quantum mechanics** and **local classical pictures**. This book's bubble / grid model belongs to **local update** intuition; after this archive you should say precisely **what statement is negated** —and **what is not**.

19.1 A. Who GHZ were and what problem they sharpened

1989, Greenberger, Horne, Zeilinger (plus Mermin's crisp version)
Bell's theorem gives a **statistical** violation for **two-body** entanglement: you need many runs to see an **average** cross of the Bell bound.

The GHZ construction shows: for **three or more bodies** in a specific entangled state (**GHZ state**), under ideal conditions you can get a **sharp logical contradiction** —

If you insist each particle has a **definite answer before measurement, independent of distant settings** (a **noncontextual, local-realistic** assignment flavor),
then a set of equalities that **should all hold collide** (a **0 = 1**-style clash).

This is the **GHZ paradox** or **GHZ theorem**: it pushes "local hidden variables assigning outcomes that way" into **single-shot** trouble, not only large-sample statistics (real experiments still have noise and imperfections).

19.2 B. GHZ state (minimal)

A three-qubit **GHZ state** is often written:

```
|GHZ = (|000 + |111) / √2
```

(Basis notation is illustrative; this book is not teaching quantum computing here.)

Intuition: not "three independent coins"; not a classical correlation you can exhaust with "if A is fixed, B follows." It is **many-body entanglement** that needs all three to describe correctly.

19.3 C. "What it proves" —three layers

Layer 1 (math–logic, ideal GHZ)
Under **standard quantum rules**, certain **commuting observable** combinations predict **perfect correlations**; if you **force** predetermined **+1 / −1 style "objective answers"** per particle (answers not depending on distant **this-round** basis choices —a common **local hidden-variable** cartoon), those answers **cannot satisfy all relations at once. So: that class of "predetermined, locally carried" assignment models is ruled out.**

Layer 2 (experiment)
Real labs approximate GHZ with ions, photons, superconducting qubits, etc.; data are **finite fidelity** —subtract noise, crosstalk, loss. The usual conclusion: **observations match standard quantum predictions** and disagree with a "simple local assignment table" —**but every sentence needs error bars and protocol.**

Layer 3 (philosophy —this book does not rule for you)
GHZ **does not** prove "consciousness creates reality" or pick an interpretation; it **hits hard** the image of **many-body entanglement as three coins with fixed answers stitched by classical correlation** —that picture **breaks** in the ideal argument.

19.4 D. Rough experimental skeleton

1. **Prepare many-body entanglement** —e.g., trapped ions steered toward |GHZ⟩ with gates.

2. **Choose measurement bases** —record **+1 / −1** or equivalent binary outcomes.

3. **Check predicted correlations** —joint probabilities vs GHZ predictions; compare to **Mermin-type inequalities** (many-body Bell bounds).

4. **Contrast with classical bounds** —if devices and protocols are trusted, violating many-body Bell / Mermin bounds parallels narratives about violating **local assignment models** (details in specialist reviews).

Difference from two-body Bell
- Bell: **statistics** decide.
- GHZ: **ideal limit** can decide on a **logic chain**; experiments still argue **finite data and noise**.

19.5 E. Directions that have been pursued (taxonomy)

Direction	Note
Ion traps	Many-ion GHZ, full gate sets, controllable fidelity and readout.
Photons	Multi-photon entanglement; loss and coincidence counting are hard; GHZ / GHZ-like reports exist.
Superconducting qubits	Many-body entanglement and "quantum advantage" style demos on chips.
Combined with Bell tests	Many-body inequalities, device-independent protocols, etc.

In this repo, `scripts/explore/ghz_medium_v10` and friends are **in-silico protocol audits** —**not** substitutes for raw lab traces —read **archive vs code separately**.

19.6 F. Why it "rules out our model" —say it cleanly

This book's model (bubble, lattice update, local coupling) **narratively** commits to:

- information / disturbance **propagates locally** cell to neighbor;
- "distant" outcomes should not depend on **instant, nonlocal coordination** unless you **explicitly add** new fields or global variables (then it is **no longer** the minimal "only local bubbles bumping bubbles" image).

What the standard GHZ argument attacks is:

- **predetermined measurement answers for each particle**;
- and answers **not "contextually" retuned when distant basis choices change this round** (otherwise you are in a different theory, not the simplest "local treasure map").

Therefore:

Statement	Ruled out by GHZ?
"Each particle has a **preprinted classical answer card**, independent of distant basis choice this round"	**Yes** —hard to save under ideal GHZ reasoning.
"Any **classical field**, any **dissipation**, any **medium**, if fancy enough, automatically escapes"	**No** —not a one-line exemption; you must **show** whether you can reproduce GHZ correlations; Chapter 3's loss **does not** hand you a blanket waiver.

Statement	Ruled out by GHZ?
"This repo's **discrete grid** has **reproduced** full standard GHZ protocol and statistics"	**No** —repo has its own bounds; see `ghz_*` script notes and failure logs.

One line to the author's model:
If you insist on **only** local updates **without** nontrivial many-body correlations at measurement events in the quantum sense, then the **many-body correlations** highlighted by standard GHZ **clash** with the **minimal classical assignment map** —that is the **precise sense of "negation"**: **negating an overly simple classical treasure map**, not automatically negating whether you might someday write **another** still-local but richer theory (that would be a new project with new experimental tests).

19.7 G. Bridge to Chapters 2 and 3

- **Chapter 2**: same data, swap **binarization yardstick**, **S** jumps —protocol sensitivity on the **Bell/CHSH** layer.
- **Chapter 3**: media **eat energy** —but **eating energy alone** does not dissolve the **GHZ assignment contradiction**; that is a **correlation-structure** problem, not only **amplitude decay**.
- **This archive**: when you say "GHZ kills the bubble model," say instead: **it kills the minimal "only local updates + simple predetermined assignments" image**; later **in-silico GHZ** asks whether **another explicit ruleset** can **approximate** the public narrative —**parallel** to this archive, **not** a substitute.

19.8 H. Deeper pointers

- Mermin's "three-coin paradox" style exposition of GHZ.
- Many-body **Bell inequalities**, **Mermin–Klyshko** bounds.
- In this book's TOC: **Chapter 11** (GHZ and in-silico protocol audit) and the `ghz_medium_v10` index.

20 第四章•Chapter 4

栅格涟漪：双缝与屏上统计
Ripple grid: double-slit and screen statistics

> **本章任务**
> 把第一章的泡泡**落实**为 `scripts/ce/` 里可跑的格子：双缝几何、屏上强度与**可见度**——并说明这些量如何随**距离与耦合**变化。

20.1 4.1 从泡泡到格子

第三章说过：**程序里每一步都会「漏」**。本章在**最干净的设定**下先看**条纹与屏统计**：`chain_explosion_numba` 内核里，能量（或强度）在二维格上按 **A、B、S** 与 **LAMBDA** 分给邻格；**一整列挡板 + 两条缝**由布尔矩阵 `barrier` 刻出来，图样是**规则推出来的**，不是后期画上去的。

20.2 4.2 `ce_00`：最小双缝故事

`scripts/ce/ce_00_double_slit_demo.py` 是入口：典型参数在脚本顶部（栅格宽高、**A/S/B**、**LAMBDA**、缝位、步数）。运行后你会得到屏上一条**累积能量曲线**与场的热力图。若侧向耦合 **S** 太小或衰减过猛，条纹会「糊」——这不是玄学，是**参数扫描**能复现的事。

20.2.1 一次运行快照（模型内，非实验室）

下面是一条**可复跑**的「档案式」记录，便于把**口头说有条纹**换成脚本里打印的键值。数值来自 **2026-05-05**、仓库根目录、**默认常量**、在终端设 `PYTHONPATH` 指向仓库根、`MPLBACKEND=Agg`；在 Windows 上另设 `PYTHONIOENCODING=utf-8` 可避免控制台在 emoji 处编码报错。**换机器或改一行参数，`local_max_count` 与 `fringe_verdict` 都可能变**——这正是本章要说的：先固定定义，再谈像不像「明显干涉」。

项目	内容
命令	在仓库根：`python scripts/ce/ce_00_double_slit_demo.py`（需能 `import chain_explosion_numba`）
常量（脚本默认）	A=1.0，S=0.25，B=0.05，λ=0.85；WIDTH=300，HEIGHT=200；STEPS=300；缝距 50px，缝宽 6px；光源 (5, HEIGHT/2)；挡板列 `BARRIER_X`=WIDTH/2
产出图	`scripts/ce/ce_00_double_slit_demo.png`（与脚本同目录）
`emit_case_dossier` 摘要	`screen_column_x`: 160；`local_max_count_on_screen`: **2**；`fringe_verdict`: `weak`；`screen_max` ≈ **7.25×10^{14}**（未归一化累积能，量级随步数变，仅作同参对比用）

脚本用右边界整列上对 `screen` 做**简单峰计数**作快速判据；默认参数下得到 `weak`，反而诚实：**栅格双缝不是「画了缝就自动 obvious」**。读者可增大 **S** 或略降 **LAMBDA** 重跑，对照 `EXPERIMENT_DOSSIER_JSON` 块里 `observed_outcome` 的变化，把叙事钉在**同一套定义**上。

20.3 4.3 `ce_01` 与可见度随距离

`ce_01_visibility_vs_screen_distance.py` 问的是：**屏放远或拉近，条纹对比（可见度）怎么变。**仓库把可见度写成屏上峰谷对比的经典式（各章若改定义会写明）。这条线把「传播几何」和**可检验曲线**绑在一起：模型若自称类波动，通常要交代**随距离的衰减或振荡**，不能只说「有缝就有纹」。

20.4 4.4 `ce_02`、`ce_03`（屏统计与侧向耦合）

- `scripts/ce/ce_02_double_slit_screen_statistics.py`：在固定双缝几何下，抽取**屏上统计量**（分布、对比等，以脚本内导出为准），便于与「肉眼判条纹」脱钩。
- `scripts/ce/ce_03_visibility_vs_side_coupling_S.py`：扫**侧向耦合 S** 对**可见度**的影响——直接回答「扩散/侧漏多了，条纹还能不能站稳」。

读法：仍用 **ce_00** 建立基线，再对照 **ce_02/ce_03** 看**换了哪几个旋钮、输出 CSV/图**落在哪。若脚本挂了 `experiment_dossier`，归档路径以运行时配置为准。

> **本章边界**
> 本章是**模型内**栅格实验；**不**引用某次真实实验室的双缝原始迹。与公开数据的对照从**后文 NIST / CHSH 章**另起炉灶。

你现在已经有一块**会漏、有缝、有屏**的**标准试验田**。下一章在同一格子上加 **缝口吸收与有限吸收体**：「测量」开始变成**局域乘子**与区域损耗——接续第三章的 **η** 语言。

21 Chapter 4 •Ripple grid: double-slit and screen statistics

栅格涟漪：双缝与屏上统计
Ripple grid: double-slit and screen statistics

> **This chapter’s job**
> Turn Chapter 1’s bubble into runnable grids in `scripts/ce/`: double-slit geometry, screen intensity, **visibility** —and show how those quantities move with **distance and coupling**.

21.1 4.1 From bubble to lattice

Chapter 3: **every step leaks.** This chapter starts in the **cleanest setting** for **fringes and screen stats**: in the `chain_explosion_numba` kernel, energy (or intensity) on a 2D grid splits to neighbors with **A, B, S** and **LAMBDA**; a **full barrier column + two slits** is a boolean `barrier` mask —the pattern is **emergent from rules**, not painted in post.

21.2 4.2 `ce_00`: minimal double-slit story

`scripts/ce/ce_00_double_slit_demo.py` is the entry: typical parameters live at the top (grid size, **A/S/B**, **LAMBDA**, slit positions, steps). You get a **cumulative energy curve** on the screen and a heatmap. If side coupling **S** is tiny or decay is brutal, fringes “mush” —not mysticism, something **parameter sweeps** reproduce.

21.2.1 One-run snapshot (in-model, not a lab)

A **rerunnable** “dossier” line replaces “we see fringes” with printed keys. Numbers from **2026-05-05**, repo root, **default constants**, terminal `PYTHONPATH` at repo root, `MPLBACKEND=Agg`; on Windows, `PYTHONIOENCODING=utf-8` avoids console emoji encoding errors. **Change machine or one constant and `local_max_count` / `fringe_verdict` may move** —the chapter’s point: **fix definitions, then argue “obvious interference.”**

Item	Content
Command	From repo root: `python scripts/ce/ce_00_double_slit_demo.py` (needs `import chain_explosion_numba`)
Constants (script defaults)	A=1.0, S=0.25, B=0.05, λ=0.85; WIDTH=300, HEIGHT=200; STEPS=300; slit spacing 50px, slit width 6px; source (5, HEIGHT/2); barrier column `BARRIER_X`=WIDTH/2
Figure	`scripts/ce/ce_00_double_slit_demo.png` (same folder as script)
`emit_case_dossier` **summary**	`screen_column_x`: 160; `local_max_count_on_screen`: **2**; `fringe_verdict`: `weak`; `screen_max` ≈ **7.25×10^{14}** (unnormalized cumulative energy —scale drifts with steps; for same-parameter comparison only)

The script uses simple peak counting on the right-edge screen column as a quick verdict; defaults yield `weak` —honestly: **grid double slit is not “obvious fringes for free.”** Raise **S** or trim **LAMBDA** and rerun; track `EXPERIMENT_DOSSIER_JSON` / `observed_outcome` so the story stays on **one definition**.

21.3 4.3 ce_01 and visibility vs distance

ce_01_visibility_vs_screen_distance.py asks: **move the screen in or out —how does fringe contrast (visibility) move?** The repo writes visibility in a classical peak–valley sense (if a chapter redefines it, the chapter says so). This ties **propagation geometry** to a **checkable curve**: a wave-like model owes a story about **decay or oscillation with distance**, not just "slits imply stripes."

21.4 4.4 ce_02, ce_03 (screen stats and side coupling)

- scripts/ ce/ ce_02_double_slit_screen_statistics.py: fixed double-slit geometry, **screen statistics** (distribution, contrast, etc. —per script exports), decoupling from "I eyeball fringes."
- scripts/ce/ce_03_visibility_vs_side_coupling_S.py: sweep **side coupling S** vs **visibility** — answers whether **more sideways leak still leaves stable fringes.**

Read ce_02/03 after ce_00 baseline: **which knobs moved, where CSV/ figures land.** If an experiment_dossier is attached, archive paths follow the run config.

> **Chapter boundary**
> This is an **in-model** grid lab; it **does not** cite raw traces from a real double-slit run. Public-data contrast starts fresh in **later NIST / CHSH chapters**.

You now have a **standard plot** that **leaks, has slits, has a screen.** Next chapter adds **slit absorption and finite absorbers** on the same lattice: "measurement" becomes **local multipliers** and regional loss —continuing Chapter 3's **η** language.

22 第五章•Chapter 5

栅格涟漪：缝口吸收、有限吸收体与探测器
Ripple grid: slit absorption and finite absorber–detector

> **本章任务**
> 把「探测器读到一点」落实为**可扫参数**：缝后吃多少、吸收片多大、多狠——看条纹与可见度**如何被洗掉。**

22.1 5.1 承接：测量 = 局域干预

专论里写过 **Ê = (1−η)E** 式的吸收。ce_04 起，脚本把 **η**、**吸收列**、**缝行区间**写进传播循环：你在传播链上**显式抠走**一部分场，再进入下一步。这里没有「口头坍缩」，只有**乘子与几何。**

22.2 5.2 ce_04：缝口吸收

scripts/ce/ce_04_measurement_absorption_at_slit.py（及同系列变体）扫描：**就在缝口**吃掉一部分能量时，下游条纹如何劣化。直觉与第三章一致：**多吞一口，就少一份能干涉的份额**；定量由曲线说话。

22.2.1 一次运行快照（ce_04 三联对照，模型内）

下列数值来自 **2026-05-05**、仓库根目录、PYTHONPATH 指向仓库根、MPLBACKEND=Agg、建议 PYTHONIOENCODING=utf-8 下的一次运行。脚本对**上缝**邻列施加分档吸收（ABSORB_RATIO 默认 **0.5**），在 SCREEN_X = WIDTH - 10 取列并计算脚本内 compute_visibility。

项目	内容
命令	`python scripts/ce/ce_04_measurement_absorption_at_slit.py`
图文件	`scripts/ce/measurement_effect.png`（三联屏曲线）

场景	`emit_case_dossier` 键	可见度 V（快照）
无吸收（双缝基线）	`visibility_none_absorb`	≈ **0.8065**
上缝 50% 吸收	`visibility_partial_absorb`	≈ **0.7719**
上缝 100% 吸收（等效单缝叙事）	`visibility_one_slit_gone`	≈ **0.8431**

读法：**部分吸收**相对基线 **V 下降**，与「吃掉相干份额」一致。第三行 **V 未跌近零**——因脚本的 **V** 是**峰谷启发式**，单缝式包络仍可有起伏；**不能**把该标量单独当「干涉是否消失」的终审，应结合 `measurement_effect.png` 与峰位对称性判断。下一节 `ce_05` 把吸收块做成**二维几何**，同一类「读数定义敏感」仍会出现。

22.3 5.3 `ce_05`：有限吸收体 / 探测器块

把吸收从「一条缝列」推广到**二维小块**：模拟**有限大探测器**或**挡在半路的海绵**。参数包括位置、宽度、**η**。你会看到：不是只有「全有或全无」，**半大不小的吸收体**就能显著改统计形状——这对后面谈「符合窗」「有效探测区」是同一类**几何敏感**。

22.3.1 一次运行快照（`ce_05` 默认圆吸收体，模型内）

下列数值来自 **2026-05-05**、仓库根目录、`PYTHONPATH` 指向仓库根、`MPLBACKEND=Agg`、建议 `PYTHONIOENCODING=utf-8` 下的一次运行。`ENABLE_ABSORBER=True`（默认）：圆心约在缝后 **(BARRIER_X+5, 上缝中心)**，**半径 10**，**ABSORB_RATIO=0.8**，**STEPS=500**，屏列 `SCREEN_X = WIDTH - 10`。

项目	内容
命令	`python scripts/ce/ce_05_finite_absorber_detector.py`
图文件	`scripts/ce/finite_absorber.png`（屏曲线 + 全场对数图 + 局部放大）
`emit_case_dossier` **摘要**	`visibility_final` = **0.0**（脚本内 `compute_visibility`：无有效峰谷时返回 0）

读法：与 `ce_04` 的「部分吃一条缝列」不同，**二维圆斑 + 高吸收** 在本组默认参下把屏上曲线压成**启发式 V 读不出来**的形态——脚本会走「对比度低于 0.1」分支。这**不**自动等价于实验室里的 which-way 结论；只说明 **几何与 η 联调** 能强烈改变**同一套 V 定义**下的输出。务必另跑 `ENABLE_ABSORBER=False` 对照，并把两幅 `finite_absorber.png`（或导出 CSV）并排归档。

> **本章边界**
> 吸收仍是**经典式损耗**在格点上的实现；**不**声称单独解决 Bell/GHZ 的赋值矛盾（见 `07-archive-ghz.md`）。

同一泡泡内核上，你多了**两把旋钮**：哪里吃、吃多狠。下一章引入**延迟插入吸收**、**测量相图**与更复杂的**纠缠扫描**脚本（`ce_06`–`ce_10`）：问「**何时**吃」与「**多种测量设定**」会怎样改变图样与相关曲线。

路标：栅格玩具段先告一段落；**Bell 与公开数据的主审计线**在**第 8–11 章**（NIST、装置拟合、CHSH 二值化、GHZ in-silico）。若你主要为那条线而来，**允许从目录直跳第 8 章**，不必把第 4–7 章当作前置考试。

23 Chapter 5 •Ripple grid: slit absorption and finite absorber–detector

栅格涟漪：缝口吸收、有限吸收体与探测器
Ripple grid: slit absorption and finite absorber–detector

> **This chapter’s job**
> Turn “the detector touches a point” into **scannable knobs**: how much is eaten past the slit, how big and how hungry the absorber —watch fringes and visibility **wash out**.

23.1 5.1 Bridge: measurement = local intervention

Reviews use **Ê = (1−η)E**-style absorption. From `ce_04` on, scripts bake **η**, **absorption columns**, **slit row ranges** into the propagation loop: you **explicitly scrape** part of the field before the next step. No verbal collapse —only **multipliers and geometry**.

23.2 5.2 `ce_04`: absorption at the slit

`scripts/ce/ce_04_measurement_absorption_at_slit.py` (and variants) scans: when you **eat energy right at the slit**, how downstream fringes degrade. Intuition matches Chapter 3: **more swallowed, less coherent share to interfere**; the curves carry the quantity.

23.2.1 One-run snapshot (`ce_04` triptych, in-model)

Numbers from **2026-05-05**, repo root, `PYTHONPATH` at root, `MPLBACKEND=Agg`, `PYTHONIOENCODING=utf-8` recommended. The script applies stepped absorption to **upper-slit** neighbor columns (`ABSORB_RATIO` default **0.5**), reads column `SCREEN_X = WIDTH - 10`, uses in-script `compute_visibility`.

Item	Content
Command	`python scripts/ce/ce_04_measurement_absorption_at_slit.py`
Figure	`scripts/ce/measurement_effect.png` (triptych)

Case	`emit_case_dossier` key	Visibility V (snapshot)
No absorption (double-slit baseline)	`visibility_none_absorb`	≈ **0.8065**
50% absorption on upper slit	`visibility_partial_absorb`	≈ **0.7719**
100% absorption on upper slit (single-slit narrative)	`visibility_one_slit_gone`	≈ **0.8431**

Readout: **partial absorption** drops **V** vs baseline —consistent with eating coherent share. The third row **does not** fall near zero —the script’s **V** is a **peak–valley heuristic**; single-slit envelopes can still undulate; **do not** treat that scalar alone as final word on “interference gone” —use `measurement_effect.png` and peak symmetry. Next, `ce_05` moves absorption into **2D geometry** — the same “readout-definition sensitivity” returns.

23.3 5.3 `ce_05`: finite absorber / detector patch

Generalize absorption from "one slit column" to a **2D patch**: finite detector or sponge mid-path. Parameters: position, width, **η**. You see it is not only all-or-nothing —**middling absorbers** can reshape statistics —the same **geometry sensitivity** that later meets **coincidence windows** and **effective sensitive areas**.

23.3.1 One-run snapshot (`ce_05` default circular absorber, in-model)

From **2026-05-05**, repo root, `PYTHONPATH`, `MPLBACKEND=Agg`, `PYTHONIOENCODING=utf-8`. `ENABLE_ABSORBER=True` (default): circle near **(BARRIER_X+5, upper-slit center)**, **radius 10**, **ABSORB_RATIO=0.8**, **STEPS=500**, screen `SCREEN_X = WIDTH - 10`.

Item	Content
Command	`python scripts/ce/ce_05_finite_absorber_detector.py`
Figure	`scripts/ce/finite_absorber.png` (screen curve + log field + zoom)
`emit_case_dossier` **summary**	`visibility_final` = **0.0** (in-script `compute_visibility`: returns 0 when no valid peaks/valleys)

Readout: unlike `ce_04`'s "partially eat one slit column," **2D blob + strong absorption** under these defaults **flattens** the screen curve so the heuristic **V** reads zero —the script takes the "contrast < 0.1" branch. This **does not** auto-equal a lab which-way verdict; it shows **geometry × η** can strongly move output under **one V definition**. Always run `ENABLE_ABSORBER=False` control and archive `finite_absorber.png` (or CSV) side by side.

> **Chapter boundary**
> Absorption is still **classical-style loss** on lattice sites; it **does not** claim to settle Bell/GHZ assignment tensions alone (see `07-archive-ghz.md`).

On the same bubble kernel you gained **two knobs**: where to eat, how hard. Next chapter adds **delayed insertion of absorption**, **measurement phase diagrams**, and richer **entanglement scans** (`ce_06`–`ce_10`): **when** you eat and **many measurement settings** —how patterns and correlation curves move.

Layer cue: This pauses the grid-toy arc; the **main Bell / public-data audit line** is **Chapters 8–11** (NIST, apparatus fit, CHSH binarization, GHZ in-silico). If that is why you came, **jump to Chapter 8** —Chapters 4–7 are not a gate exam.

24 第六章•Chapter 6

栅格涟漪：延迟选择、测量相图与纠缠扫描
Ripple grid: delayed choice, measurement phase diagram, entanglement scans

> **本章任务**
> 在格子上讨论：**吸收/分支规则晚一点插进去**，图样是否变；以及在多参数平面上**系统扫**「测量强度 × 耦合 × .....」时，可见度、对比度等**相图状**结构。

24.1 6.1 延迟选择：`ce_06`

「延迟选择」在科普里常被神秘化。在仓库里，它更接近工程问题：**第几步**开始启用吸收掩模？

- `scripts/ce/ce_06_delayed_choice_absorber.py`：把**吸收体插入步数**参数化；早插 vs 晚插，下游干涉条件不同则屏上统计应**可分辩**——**因果顺序写进循环**，不是口号。

24.1.1 一次运行快照（默认参，模型内）

下列数值来自 **2026-05-05**、仓库根目录、`PYTHONPATH` 指向仓库根、`MPLBACKEND=Agg`、建议 `PYTHONIOENCODING=utf-8`（Windows 控制台）下的一次运行。**吸收圆**在第 **300** 步插入（总步数 **500**），盖住**上缝**下游邻域；屏列为 `SCREEN_X = WIDTH - 10`。

项目	内容
命令	`python scripts/ce/ce_06_delayed_choice_absorber.py`
常量（脚本默认摘录）	HEIGHT=201，WIDTH=400；A=1.0，S=0.4，B=0.05，λ=0.90；INSERT_STEP=300；ABSORB_RATIO=0.9；吸收圆心约 (BARRIER_X+5, 上缝中心)，半径 8
`emit_case_dossier` **摘要**	`visibility_final` ≈ **0.8065**（脚本内 `compute_visibility(screen)`，屏列能量）
图文件	`scripts/ce/delayed_choice.png`（与 `ce_00` 相同，写入脚本所在目录，**不**依赖当前工作目录）

读法：本次默认参下 **V 仍高**，对应「晚插圆吸收**未**把条纹打到脚本里的 `<0.1` 判据」——**不**等于量子实验结论；只说明 **同一套可见度定义下，几何与步数可复算**。对比 `INSERT_STEP=0` 与「无吸收器」应另跑对照，并把三条曲线的 **V** 并排写进笔记。

24.2 6.2 测量相图：`ce_07`

- `scripts/ce/ce_07_measurement_phase_diagram_scan.py`：在参数平面上扫**测量强度、位置、耦合**等，输出「相图」式摘要（条纹活/死区域）。读法：**固定内核与几何，只动测量旋钮**——与第二章「换尺子」同精神。

24.3 6.3 `ce_08`–`ce_10`：分支、相位与距离扫描

- `scripts/ce/ce_08_entanglement_split_wavepackets.py`：分束/分支掩模下的**双路包络**与相关统计入口。
- `scripts/ce/ce_09_entanglement_with_phase_field.py`：在格点上引入**相位场**与纠缠读出（接口见脚本顶部）。
- `scripts/ce/ce_10_entanglement_distance_scan.py`：扫**空间分离/等效距离**对相关量的影响，为 **第七章** `ce_bell*` 搭桥。

具体数据结构以各脚本内 `split_mask`、**相位网格、输出数组** 为准。

> **本章边界**
> 格点上的「纠缠扫描」是**玩具相关矩阵**在定义下的行为；**不**自动等同于某实验室申报的纠缠见证。

你现在能问：**测量规则何时介入、介入多狠、在参数平面上处于何区**——全是可复算问题。下一章把这些机制**接到 Bell/CHSH 标量与前沿曲线**（`ce_bell*`）。

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25 Chapter 6 •Ripple grid: delayed choice, measurement phase diagram, entanglement scans

栅格涟漪：延迟选择、测量相图与纠缠扫描
Ripple grid: delayed choice, measurement phase diagram, entanglement scans

> **This chapter's job**
> On the lattice: **if absorption / branching rules are inserted late**, do patterns change; and on multi-parameter planes, **systematically scan** "measurement strength × coupling × ..." for **phase-diagram-like** structure in visibility and contrast.

25.1 6.1 Delayed choice: `ce_06`

Pop accounts mystify "delayed choice." In the repo it is closer to engineering: **at which step** does an absorbing mask turn on?

- `scripts/ce/ce_06_delayed_choice_absorber.py`: parameterize **insertion step** for the absorber; early vs late insert changes downstream interference —**causal order lives in the loop**, not in slogans.

25.1.1 One-run snapshot (defaults, in-model)

2026-05-05, repo root, `PYTHONPATH`, `MPLBACKEND=Agg`, `PYTHONIOENCODING=utf-8` (Windows console). **Absorbing disk** inserts at step **300** (total steps **500**), covers **upper-slit** downstream neighborhood; screen column `SCREEN_X = WIDTH - 10`.

Item	Content
Command	`python scripts/ce/ce_06_delayed_choice_absorber.py`
Constants (snippet)	HEIGHT=201, WIDTH=400; A=1.0, S=0.4, B=0.05, λ=0.90; INSERT_STEP=300; ABSORB_RATIO=0.9; disk center ~(BARRIER_X+5, upper-slit center), radius 8
`emit_case_dossier` **summary**	`visibility_final` ≈ **0.8065** (`compute_visibility(screen)` on screen column energy)
Figure	`scripts/ce/delayed_choice.png` (same pattern as `ce_00`: written next to the script, **not** cwd-dependent)

Readout: under these defaults **V stays high** —late disk **does not** push fringes below the script's `<0.1` test —**not** a quantum verdict; only that **geometry × step index** recomputes under **one visibility definition.** Contrast `INSERT_STEP=0` vs "no absorber" in separate runs and tabulate **V**.

25.2 6.2 Measurement phase diagram: `ce_07`

- `scripts/ce/ce_07_measurement_phase_diagram_scan.py`: scan **measurement strength, position, coupling** on a parameter plane; output "phase-map" summaries (live/dead fringe regions). **Fix kernel and geometry, move only measurement knobs** —same spirit as Chapter 2's "swap yardsticks."

25.3 6.3 ce_08–ce_10: branching, phase field, distance scan

- `scripts/ce/ce_08_entanglement_split_wavepackets.py`: dual-path envelopes and correlation stats under split / branch masks.
- `scripts/ce/ce_09_entanglement_with_phase_field.py`: phase field on the grid and entanglement-style readout (see script header).
- `scripts/ce/ce_10_entanglement_distance_scan.py`: sweep **spatial separation / effective distance** for correlation metrics —bridge to **Chapter 7** `ce_bell*`**.**

Data layouts follow each script' s `split_mask`, phase grid, output arrays.

> **Chapter boundary**
> Grid "entanglement scans" are **toy correlation matrices** under stated definitions; they **do not** auto-equal a lab' s entanglement witness.

You can now ask: **when measurement rules bite, how hard, and where in parameter space** —all recomputable. Next chapter pipes these mechanisms into **Bell/CHSH scalars and frontier curves** (`ce_bell*`).

26 第七章•Chapter 7

栅格 Bell/CHSH 玩具与前沿曲线
Grid Bell/CHSH toys and frontier curves

> **本章任务**
> 说明仓库如何用**同一套局域传播 + 分支/偏振规则**构造 **CHSH 型统计**；**S** 或等价量在参数空间里如何扫出**前沿**——以及这些结果**只**在脚本定义下有效。

26.1 7.1 为什么需要「玩具」层

第二章的 **1.1 / 2.3** 来自**公开记录 + 后处理**。在此之前，仓库用 `ce_bell*` 在**完全可控**的格子上问：**当耦合、噪声、阈值、分束比变化时，类 CHSH 指标怎么走。**这是**模型内实验室**：协议写在你眼皮底下。

26.2 7.2 ce_bell* 脚本索引（仓库快照）

下列均在 `scripts/ce/`；职责以文件名为纲，细节以**脚本内协议块**为准：

文件	大致用途
`ce_bell.py`	Bell/CHSH **栅格玩具**基线入口之一
`ce_bell_chsh_evaluator.py`	CHSH **评估器**（计数 →S 或等价量，定义锁定在文件内）
`ce_bell_chsh_frontier.py`	参数空间 **前沿曲线**扫描
`ce_bell_chsh_robustness_fixed.py`	**鲁棒性**对照（固定协议下的扰动）
`ce_bell_final_curve_fixed.py`	**定版曲线**复现/汇总型脚本
`ce_bell_v11_stress_test.py`	**应力测试**（缺陷图、噪声等）

典型要素仍是：两路或多路、**二元标签/偏振代理**、在 **A、S、B、LAMBDA** 上扫描。**不同脚本的「事件」与分母可能不同**——比较两次运行前，先**对齐定义。**

26.2.1 一次运行快照（ce_bell.py：马吕斯曲线 vs cos²）

ce_bell.py 对 **10 个偏振角差** 各跑一整轮 **STEPS=800** 传播，在固定探测器像素上取 **符合量 ~ m1×m2**，再除以 **angle2=0** 处的基线做归一化，与 **cos²θ** 逐点比（定义以脚本为准）。图写入 scripts/ce/bell_test_result.png。

项目	内容
命令	在仓库根：PYTHONPATH 指向仓库根后执行 python scripts/ce/ce_bell.py
快照日期	**2026-05-05**（脚本默认 HEIGHT/WIDTH、A/S/B、λ、**ANGLES** 等）
基线符合（未归一化）	约 **3.79×10□□**（angle2=0 时的 coincidence，仅作归一化分母）
对 cos² 的 RMSE	约 **4.5×10□¹⁸**（双精度下与解析曲线几乎重合）
np.allclose(..., atol=0.1)	**True**（脚本自检；**仅**曲线贴合，**不是**完整 Bell/CHSH 认证——见脚本 log **「not a full Bell-test certification」**）

读法：这一层是 **「玩具符合 + 马吕斯代理」**；**CHSH 数值与 NIST 档案**在 **第八至十章** 另起炉灶。勿把本图的 **cos² 贴合** 说成实验室 **S 值** 或 **无局域性** 结论。

26.3 7.3 前沿与 stress

前沿：见 ce_bell_chsh_frontier.py。**应力**：见 ce_bell_v11_stress_test.py（缺陷格、噪声式踢脚等）。目的：**模型在受挤时是否自洽**，不是替自然界下判决书。

> **本章边界**
> 玩具层 ≠ NIST 审计层；前者服务**直觉与灵敏度**，后者服务**公开记录上的尺子**（第八至十章）。**禁止**把某次 ce_bell 的数值直接说成「实验室已证伪/证实」。

你有了**从双缝可见度走到 CHSH 形状**的**内部桥梁**。下一章离开纯玩具，进入 **NIST complete-blind 类公开档案**：**分母、符合窗、索引语义**成为主角。

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27 Chapter 7 •Grid Bell/CHSH toys and frontier curves

栅格 Bell/CHSH 玩具与前沿曲线
Grid Bell/CHSH toys and frontier curves

> **This chapter’s job**
> Explain how the repo builds **CHSH-style statistics** from **one local propagation stack + branching / polarization rules**; how **S** (or an equivalent) is scanned into **frontiers** in parameter space —and why these numbers are **only** valid under each script’s definition.

27.1 7.1 Why a “toy” layer exists

Chapter 2’s **1.1 / 2.3** came from **public records + post-processing**. Before that, the repo uses ce_bell* on a **fully controlled** lattice to ask: **as coupling, noise, threshold, and split ratio move, how do CHSH-like figures behave?** This is an **in-model lab**: the protocol is written where you can see it.

27.2 7.2 `ce_bell*` script index (repo snapshot)

All under `scripts/ce/`; details follow the **protocol blocks inside each script**.

File	Rough role
`ce_bell.py`	Baseline **grid toy** entry for Bell/CHSH
`ce_bell_chsh_evaluator.py`	CHSH **evaluator** (counts → S or equivalent; definition locked in file)
`ce_bell_chsh_frontier.py`	**Frontier** scans in parameter space
`ce_bell_chsh_robustness_fixed.py`	**Robustness** checks under a fixed protocol
`ce_bell_final_curve_fixed.py`	**Frozen-curve** reproduce / summary style
`ce_bell_v11_stress_test.py`	**Stress test** (defect grids, noise kicks, etc.)

Typical ingredients: two or more paths, **binary tags / polarization proxies**, scans over **A, S, B, LAMBDA**. **Different scripts may define "events" and denominators differently** —align definitions before comparing two runs.

27.2.1 One-run snapshot (`ce_bell.py`: Malus curve vs cos²)

`ce_bell.py` runs **10 polarization angle gaps**, each for **STEPS=800** propagation, takes **coincidence ~ m1×m2** on fixed detector pixels, normalizes by the baseline at **angle2=0**, and compares pointwise to **cos²θ** (definition per script). Figure: `scripts/ce/bell_test_result.png`.

Item	Content
Command	From repo root with `PYTHONPATH` set: `python scripts/ce/ce_bell.py`
Snapshot date	**2026-05-05** (script defaults for HEIGHT/WIDTH, A/S/B, λ, **ANGLES**, etc.)
Baseline coincidence (unnormalized)	about **3.79×10□□** (at `angle2=0`, normalization denominator only)
RMSE vs cos²	about **4.5×10□¹⁸** (nearly exact in double precision)
`np.allclose(..., atol=0.1)`	**True** (script self-check; **only** curve agreement, **not** full Bell/CHSH certification —see log **"not a full Bell-test certification"**)

How to read this: This layer is **"toy coincidence + Malus proxy"**; **CHSH values and NIST archives** start in **Chapters 8–10**. Do not sell this plot's **cos² fit** as a lab **S** or a **no-locality** conclusion.

27.3 7.3 Frontiers and stress

Frontiers: `ce_bell_chsh_frontier.py`. **Stress**: `ce_bell_v11_stress_test.py` (defected grids, noise-like kicks). Goal: **whether the model stays coherent under pressure**, not a verdict on nature.

> **Chapter boundary**
> Toy layer ≠ NIST audit layer; the former serves **intuition and sensitivity**, the latter **yardsticks on public records** (Chapters 8–10). **Do not** relabel a `ce_bell` number as "the lab has falsified / confirmed ..."

You now have an **internal bridge from double-slit visibility to CHSH-shaped curves**. Next chapter leaves pure toys and opens **NIST complete-blind-style public archives**: **denominators, coincidence windows, index semantics** take the lead.

28 第八章•Chapter 8

公开数据：NIST complete-blind 上的 Bell/CHSH 分母与符合窗审计
Public data: Bell/CHSH denominator and coincidence-window audit on NIST complete-blind

> **本章任务**
> 把叙事从格子玩具**切到磁盘上的 CSV**：同一批 **NIST** 公开记录，**分母怎么取、符合怎么配对、窗宽怎么选**——会如何移动 **S 或 CHSH 代理量。**

28.1 8.1 为什么必须单独开一章

第二章只给你**一个戏剧性的数差。**本章（及后续数章）要做的是：**把戏剧拆成步骤**——哪一步对应仓库里哪条审计脚本、哪条归档 JSON。**complete-blind** 类档案有其**字段语义**；若把索引或权重误读，统计对象就悄悄换了。

28.2 8.2 分母与「有效对」：仓库从哪下手

Bell/CHSH 不是只有一个公式外壳，内里常藏：**哪些 trial 计入、哪些因丢失或无效被扔掉。**本书仓库在下列线索上**反复审计**（非穷尽列表，随仓库演进会增删）：

方向	示例脚本（`scripts/explore/`）
字段/索引语义统一	`nist_unified_semantics_audit_v1.py` …`v4.py`
同一索引上的二值化/量化扫描	`nist_same_index_quantization_sweep_v1.py`、`nist_same_index_quantization_sweep_v4.py`
ρ–μ–η 或双指标桥	`nist_same_index_rho_mu_eta_scan_v1.py`、`nist_same_index_dual_metric_bridge_v1.py`
角度符号与符合构造	`nist_same_index_angle_sign_scan_v1.py`
closure / 复盘包	`nist_revival_20pct_closure_v1.py`、`nist_revival_20pct_closure_v4.py`
邮件/摘要打包	`nist_email_summary_pack_v4.py`、`nist_semantics_contribution_summary_v1.py`

归档常见根：`battle_results/`、`papers_final/`（以你本地克隆为准）。读法：**先打开具体 `*_audit*.py` 顶部的「事件定义」**，再对照 **CSV 列名。**

28.3 8.3 符合窗、时钟与「时间对齐」

光子实验中，**符合窗**决定哪些计数算「同时」；窗宽一变，偶然符合与真符合比例变——**S 会跟着动。**仓库中亦有**时钟参考/对齐**类审计（如 `nist_clock_reference_audit_v1.py`），用于检查**时间轴语义**是否与宣称一致。要点仍是：**把假设写进脚本**，再让数字说话。

> **本章边界**
> 本章是**对公开记录与协议的审计式讨论**；**不是**对 NIST 或实验者的人身评价。任何 headline 数字以**本章引用的归档文件**中的定义为准。

你应带走：**公开数据章节的硬度，来自分母与窗的显式化。**下一章进入**更几何的独立实验叙事**：在**同批数据**上拟合 Alice/Bob 偏振路径的**等效庞加莱坐标**——又一层「尺子」，可与 CHSH 管道**交叉检验。**

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29 Chapter 8 •Public data: Bell/CHSH denominator and coincidence-window audit on NIST complete-blind

公开数据：NIST complete-blind 上的 Bell/CHSH 分母与符合窗审计
Public data: Bell/CHSH denominator and coincidence-window audit on NIST complete-blind

> **This chapter’s job**
> Move the story from lattice toys **to CSV on disk**: on one **NIST** public record, **how denominators are chosen, how coincidences pair, how window width is set** —and how those choices move **S** or a **CHSH proxy**.

29.1 8.1 Why this deserves its own chapter

Chapter 2 only showed a **dramatic gap**. This chapter (and neighbors) **unpack the drama into steps** —which step maps to which audit script, which archived JSON. **complete-blind** archives have **field semantics**; misread an index or weight and the statistical object quietly changes.

29.2 8.2 Denominators and “effective pairs”: where the repo digs in

Bell/CHSH is not just a formula shell; inside hides **which trials count, which are dropped for loss or invalidity**. The repo revisits these threads (non-exhaustive; evolves with the repo):

Direction	Example scripts (`scripts/explore/`)
Unified field / index semantics	`nist_unified_semantics_audit_v1.py` …`v4.py`
Same-index binarization / quantization sweeps	`nist_same_index_quantization_sweep_v1.py`, `nist_same_index_quantization_sweep_v4.py`
ρ–μ–η or dual-metric bridge	`nist_same_index_rho_mu_eta_scan_v1.py`, `nist_same_index_dual_metric_bridge_v1.py`
Angle sign and coincidence construction	`nist_same_index_angle_sign_scan_v1.py`
Closure / review packs	`nist_revival_20pct_closure_v1.py`, `nist_revival_20pct_closure_v4.py`
Email / summary bundles	`nist_email_summary_pack_v4.py`, `nist_semantics_contribution_summary_v1.py`

Archives often live under `battle_results/`, `papers_final/` (per your clone). Reading discipline: **open the chosen `*_audit*.py`, read the “event definition” header**, then line up **CSV column names**.

29.3 8.3 Coincidence windows, clocks, and “time alignment”

In photon work, the **coincidence window** decides which counts are “simultaneous”; change the width and the accidental vs true ratio shifts —**S moves with it**. The repo also has **clock reference / alignment** audits (e.g. `nist_clock_reference_audit_v1.py`) to check whether **time-axis semantics** match what is claimed. Same moral: **write assumptions into the script**, then let numbers speak.

> **Chapter boundary**
> This is **audit-style discussion of public records and protocols** —**not** a personal judgment on NIST or the experimenters. Any headline number follows the **definition in the archived file cited in this chapter**.

Take away: **public-data chapters get their rigor from explicit denominators and windows.** Next chapter opens a **more geometric, standalone experiment narrative**: fit **effective Poincaré coordinates** for Alice/Bob polarization paths on **the same batch** —another “yardstick”that can **cross-check** the CHSH pipeline.

30 第九章•Chapter 9

基于 NIST 数据：逐次测量拟合偏振装置在庞加莱球上的等效坐标
NIST-based experiment: per-trial fit of effective Poincaré-sphere coordinates

> **本章任务**
> 说明仓库中**独立于 CHSH 主线的拟合实验**：把单次 trial 的读出，放到**等效光学元件**（庞加莱球上的角度/相位）语言里，检验**可辨识性**与**不确定度预算。**

30.1 9.1 为什么要「另一条尺子」

CHSH 关心**关联是否超界。**拟合路线问：**若把装置当成黑箱，单次数据更支持哪组等效参数？**两条线**共享同一 CSV**，但**统计对象不同**——这正是序言里「分开写清」的实例。

30.2 9.2 逐次拟合与「几何参数」：从哪份脚本进门

本章标题里的**庞加莱球**，在工程上对应：**偏振态 / 波片角 / 等效相位**的一组**连续参数。**仓库**不**保证只有一个「官方拟合脚本」；常见进门点包括：

- `scripts/explore/nist_same_index_angle_sign_scan_v1.py`：**角度符号**与指数定义对统计的敏感扫描——拟合前先要**符号约定**不打架。
- `scripts/explore/explore_nist_e_delta_three_tracks.py`：同一底层表上 **多轨（tracks）**处理对照（文档串里可见 **Alice/Bob one-hot** 等有效样本定义）——与 CHSH 轨**并行**时，须锁**同一 trial 子集。**

对每次符合事件（定义见**你锁定的**那份脚本），构造**代价函数**，在 **Alice/Bob** 各一组角度上搜索最优。输出应包含：**点估计**、**是否病态**（多解、平坦方向）、**不确定度**（自助法或局部线性近似）。**没有不确定度而只有点估计的图，本书不当作定论。**

30.3 9.3 可辨识性与诚实失败

若数据对某参数组合**几乎不敏感**，拟合会给出**大不确定度**或**多谷**——这在工程上叫**不可辨识**或**弱辨识。**本书把这类输出也当作**有效结果**：说明「这条尺子在此数据上**不够锋利**」，而不是强行给漂亮点估计。

> **本章边界**
> 等效坐标是**模型类内**的参数化；**不**自动等于设备商提供的校准证书。与 CHSH 数值的**交叉引用**须在**同一 trial 子集与权重**前提下进行。

与第十五章（规划稿）的关系：**第 15 章**预留 **Bell 类公开数据上 Alice/Bob 球面坐标联合拟合** 的专章——**文稿占位、代码待作者有空再写**；**本章已述语言与纪律仍为准**，将来实现时应与此章**对齐定义**而非另起炉灶。

你现在有两类对象：**Bell 型标量**与**装置几何型参数。**下一章专门把 **CHSH** 再拆：**二值化轨道 vs 连续代理轨道**，并用**合成样本**做**阴性对照。**

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31 Chapter 9 •NIST-based experiment: per-trial fit of effective Poincaré-sphere coordinates

基于 NIST 数据：逐次测量拟合偏振装置在庞加莱球上的等效坐标
NIST-based experiment: per-trial fit of effective Poincaré-sphere coordinates

> **This chapter’s job**
> Describe **fitting experiments independent of the CHSH headline pipeline**: put per-trial

readouts in the language of **effective optical elements** (angles / phases on the Poincaré sphere), and test **identifiability** plus an **uncertainty budget**.

31.1 9.1 Why a "second yardstick" helps

CHSH asks whether **correlations cross a bound**. The fitting line asks: **treating the apparatus as a black box, which equivalent parameters does each trial favor?** Both lines **share one CSV** but **different statistical objects** —the preface's "write them separately" in practice.

31.2 9.2 Per-trial fitting and "geometry parameters": where to enter the repo

Poincaré sphere here means a **continuous parameter block** for **polarization / waveplate angles / effective phase**. The repo does **not** promise a single "official fit script"; common entry points:

- `scripts/explore/nist_same_index_angle_sign_scan_v1.py`: sensitivity of statistics to **angle sign** and index definitions —**fix sign conventions before fitting**.
- `scripts/explore/explore_nist_e_delta_three_tracks.py`: **multi-track** handling on the same underlying table (strings mention **Alice/Bob one-hot** style valid-sample definitions) —when run **parallel** to CHSH tracks, **lock the same trial subset**.

For each coincidence event (definition per **your chosen** script), build a **cost function** and search optimal angles for **Alice** and **Bob**. Outputs should include **point estimates**, **ill-conditioning** (multiple solutions, flat directions), and **uncertainty** (bootstrap or local linearization). **Plots with point estimates but no uncertainty are not treated as verdicts here.**

31.3 9.3 Identifiability and honest failure

If the data are **nearly insensitive** to some parameter combo, the fit returns **huge uncertainty** or **multiple valleys** —**unidentifiable** or **weakly identifiable** in engineering terms. This book treats that as a **valid result**: the yardstick is **not sharp enough on this table**, rather than forcing a pretty point estimate.

> **Chapter boundary**
> Equivalent coordinates are **parameterizations inside a model class**; they do **not** automatically equal vendor calibration sheets. **Cross-talk with CHSH numbers** must share the **same trial subset and weights**.

Relation to Chapter 15 (planning draft): Chapter **15** reserves a dedicated slot for **joint Alice/Bob sphere-coordinate fitting on Bell-type public data** —**manuscript placeholder, code when the author has bandwidth**; **the language and discipline in this chapter remain authoritative**; future code should **align definitions** here rather than invent a parallel dialect.

You now have two kinds of objects: **Bell-style scalars** and **device-geometry parameters**. Next chapter unpacks **CHSH** again: **binarization track vs continuous-proxy track**, with **synthetic negative controls**.

32 第十章•Chapter 10

CHSH：二值化、连续代理与统计对象分轨
CHSH: binarization, continuous proxies, and split statistical objects

> **本章任务**
> 正面写清第二章的跳跃：**同一底层表**，如何分叉成 **±1 硬标签**与**保留连续幅度**两轨；为何 **S** 会「抬栏杆」；**合成数据**如何检验脚本是否**自洽**。

32.1 10.1 二值化：一把削脚的尺子

把连续电压或计数映射成 **+1/-1**，阈值、增益、饱和都会进入**有效定义**。仓库用**固定规则**重复全流程，让你看到：**不是「物理变了」，是「统计对象变了」。**

32.2 10.2 连续代理与「严格 vs 后选」对照

另一轨保留**连续量**或**更严格的符合定义**，构造**不等价但可比**的 CHSH 代理。仓库中 `scripts/explore/explore_chsh_*.py` 族覆盖多种协议切片，例如：

- `explore_chsh_user_discrete_protocol.py`：用户侧**离散协议**与 CHSH 管道对齐实验。
- `explore_chsh_strict_protocol.py`、`explore_chsh_strict_vs_postselected_compare.py`：**严格集** vs **后选**集对照。
- `explore_chsh_operation_audit.py`、`explore_chsh_experiment_alignment.py`：**操作/对齐**审计。
- `explore_chsh_wavefront_local_v23.py`、`explore_chsh_wavefront_global_rephase_v24.py`：**波前/重相位**敏感性与闭包类讨论。

分母专审：`scripts/explore/chsh_denominator_audit.py`——直接对准「**分母是谁**」这一问。

与 `nist_same_index_quantization_sweep_*.py` 并排读：看**同一索引行**上，**量化规则**如何移动 **S**。

32.3 10.3 合成样本与阴性对照

`nist_unified_semantics_audit_v1.py` 等文件内含 **synthetic_cos_events** 一类**合成事件**构造：用已知分布注入管道，若仍算不出自洽界，优先怀疑**实现**而非「Nature 错了」。阴性对照是**工程纪律**，不是修辞。

> **本章边界**
> 本章**不**裁决「量子纠缠是否存在」；只展示：**协议选择移动统计对象**——与 Bell 精神（假设写清）一致。

尺子分轨之后，公开数据的叙述才能**不糊在一起**。下一章转向 **GHZ**：**档案里的多体矛盾** vs 仓库 `ghz_medium_v10` 的 **in-silico 协议审计**。

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33 Chapter 10 •CHSH: binarization, continuous proxies, and split statistical objects

CHSH：二值化、连续代理与统计对象分轨
CHSH: binarization, continuous proxies, and split statistical objects

> **This chapter’s job**
> Spell out Chapter 2’s jump: on **one underlying table**, how pipelines fork into **hard ±1 labels** vs **keeping continuous amplitude**; why **S** can “raise the bar” ; how **synthetic data** tests **internal consistency**.

33.1 10.1 Binarization: a foot-chopping ruler

Mapping continuous voltage or counts to **+1/-1** pulls threshold, gain, and saturation into the **effective definition**. The repo reruns the full flow under **fixed rules** so you can see: **physics did not change —the statistical object did.**

33.2 10.2 Continuous proxies and "strict vs post-selected" contrasts

The other track keeps **continuous quantities** or **stricter coincidence definitions**, building **CHSH proxies that are not equivalent but comparable**. The `scripts/explore/explore_chsh_*.py` family covers many protocol slices, e.g.:

- `explore_chsh_user_discrete_protocol.py`: user-side **discrete protocol** aligned with CHSH plumbing.
- `explore_chsh_strict_protocol.py`, `explore_chsh_strict_vs_postselected_compare.py`: **strict set** vs **post-selected** set.
- `explore_chsh_operation_audit.py`, `explore_chsh_experiment_alignment.py`: **operation / alignment** audits.
- `explore_chsh_wavefront_local_v23.py`, `explore_chsh_wavefront_global_rephase_v24.py`: **wavefront / rephase** sensitivity and closure-style discussion.

Denominator-focused audit: `scripts/explore/chsh_denominator_audit.py` —asks outright **"who is the denominator?"**

Read alongside `nist_same_index_quantization_sweep_*.py`: on **the same index rows**, watch how **quantization rules** move **S**.

33.3 10.3 Synthetic samples and negative controls

Files like `nist_unified_semantics_audit_v1.py` include **synthetic_cos_events**-style **synthetic event** builders: inject a known distribution; if the pipeline still cannot recover a coherent bound, suspect **implementation** before "Nature is wrong." Negative controls are **engineering discipline**, not rhetoric.

> **Chapter boundary**
> This chapter does **not** rule on "whether entanglement exists"; it shows **protocol choice moves the statistical object** —aligned with Bell's spirit (assumptions explicit).

Once **yardsticks are forked**, public-data stories stop **smearing together**. Next chapter turns to **GHZ**: **many-body logic in the archive** vs `ghz_medium_v10` **in-silico protocol audit** in the repo.

34 第十一章•Chapter 11

GHZ 与 in-silico 协议审计
GHZ and in-silico protocol audit

> **本章任务**
> 分清三件事：**标准 GHZ 叙事**（见 `07-archive-ghz.md`）、仓库 **in-silico** 里**门控/保留率/介质 toy** 在干什么、以及**为什么不能**用本脚本输出替代实验室原始迹。

34.1 11.1 档案 vs 代码

档案回答：**局域经典赋值为何在理想 GHZ 论证里碰壁。**`scripts/explore/ghz_medium_v10/` 等回答：**在给定离散规则与协议下，我们能跑到哪一步、哪里失败、哪里需要加成本曲线。**两者**互补**，**不**互替。

34.2 11.2 `ghz_medium_v10` 族：协议对象是什么

目录 `scripts/explore/ghz_medium_v10/`（仓库快照）：

文件	大致用途
`explore_ghz_medium_wave_v10.py`	**v10 波/介质**主探索入口
`v10_1_metric_scalpel.py`	**指标解剖**（scalpel）
`v10_2_metric_robustness.py`	**鲁棒性**对照
`v10_3_selection_tax_audit.py`	**选样/税**（selection tax）审计——分母是否被「税」改写
`v10_4_real_cost_curve.py`	**成本曲线**（保留率、门控代价等）

典型产物：**阈值扫描**、**保留率**、**门控子样本**、与**预设失败条件**的对照。读报告时先找：**统计的是哪些事件**、**分母是否被筛选重写**——与第八至十章同一纪律。

34.3 11.3 其他 `ghz_*` 审计与失败归档

并行可读（**非** `medium_v10` 独占）：`ghz_chain_explosion_audit_v16.py`、`ghz_hardcore_audit_v18.py`、`ghz_phase_loop_cost_curve_v14.py`、`ghz_honest_cost_curve_v11.py` 等——命名里带 **honest / hardcore / cost** 的，往往**刻意**把**难看曲线**也存档。

仓库刻意保留**未闭合**与**红色自检**：防止叙事跑在数据前面。

> **本章边界**
> **in-silico GHZ** 是**模型内**探索；**不**构成对具体实验室 GHZ 实验的复刻结论。headline 数字以 `artifacts/ghz_*` 或章内引用的 JSON 为准。

多体线是**硬**的：直觉上泡泡再美，也要过 **GHZ 类相关**这一关——档案说明**关卡的逻辑形状**，代码说明**我们离那形状还有多远**。下一章换一条**并行管线**：**涟漪量子**四项玩具基准（激光、半导体、MRI、原子钟）。

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35 Chapter 11 •GHZ and in-silico protocol audit

GHZ 与 in-silico 协议审计
GHZ and in-silico protocol audit

> **This chapter’s job**
> Separate three things: **standard GHZ narrative** (see `07-archive-ghz.md`), what **in-silico gating / retention / medium toys** do in the repo, and **why** script output cannot stand in for laboratory raw traces.

35.1 11.1 Archive vs code

The archive answers **why naive local classical assignments hit a wall in ideal GHZ reasoning**. `scripts/explore/ghz_medium_v10/` and friends answer **under stated discrete rules and protocols, how far we can run, where we fail, where cost curves must be added**. They **complement** each other; they do **not** substitute.

35.2 11.2 The `ghz_medium_v10` family: what statistical object?

Directory `scripts/explore/ghz_medium_v10/` (repo snapshot):

File	Rough role
`explore_ghz_medium_wave_v10.py`	Main **v10 wave / medium** exploration entry
`v10_1_metric_scalpel.py`	**Metric scalpel**

File	Rough role
`v10_2_metric_robustness.py`	**Robustness** checks
`v10_3_selection_tax_audit.py`	**Selection-tax** audit —whether denominators are rewritten by "tax"
`v10_4_real_cost_curve.py`	**Cost curves** (retention, gating price, etc.)

Typical artifacts: **threshold scans**, **retention**, **gated subsamples**, contrasts with **preset failure conditions**. When reading reports, first find **which events are tallied** and **whether denominators are rewritten by selection** —same discipline as Chapters 8–10.

35.3 11.3 Other `ghz_*` audits and failure archives

Also readable (not exclusive to `medium_v10`): `ghz_chain_explosion_audit_v16.py`, `ghz_hardcore_audit_v18.py`, `ghz_phase_loop_cost_curve_v14.py`, `ghz_honest_cost_curve_v11.py` —names with **honest / hardcore / cost** often **intentionally** archive **ugly curves** too.

The repo keeps **open loops** and **red self-checks** on purpose: stop narrative from outrunning data.

> **Chapter boundary**
> **in-silico GHZ** is **in-model** exploration; it is **not** a replication verdict on any specific lab GHZ run. Headline numbers follow `artifacts/ghz_*` or JSON cited in-chapter.

The many-body thread is **hard**: however pretty the bubble intuition, it still faces **GHZ-style correlations** —the archive states **the logical shape of the hurdle**, the code states **how far we still are from that shape**. Next chapter switches to a **parallel pipeline**: the **ripple-quantum** four toy benchmarks (laser, semiconductor, MRI, atomic clock).

36 第十二章•Chapter 12

涟漪量子四项玩具基准 v1–v8
Ripple quantum four benchmarks v1–v8

> **本章任务**
> 介绍 `scripts/explore/ripple_quantum_tests/`：**四条一维曲线**上，用**无量纲介质旋钮**去贴 **QM-like 参考形状**；版本 **v1–v8** 各解决什么问题（门禁、色散、合并页等）。

36.1 12.1 为什么与 `ce_*` 并行

`ce_*` 是**二维格 + 双缝/Bell** 主战场；涟漪量子线是**另一套**参数化介质 toy，服务**四条独立「行业曲线」**直觉：阈值、截止、拉莫尔线性、谱线形。它**不**替代 NIST 事件表，也**不**声称 SI 已全局标定。

36.2 12.2 四条曲线分别是什么

- **激光**：阈值附近功率–泵浦形状。
- **半导体**：吸收边附近特征。
- **MRI**：拉莫尔关系与代数 **κ**（版本靠后引入）。
- **原子钟**：谱线与腔长代数（版本演进见索引）。

36.3 12.3 版本演进（v1→v8）与权威索引

唯一导航入口（仓库内）：`docs/RIPPLE_QUANTUM_TESTS_INDEX.md`（中英并列、含一键命令与 `artifacts/` 目录说明）。中文任务细节：`scripts/explore/ripple_quantum_tests/README.zh.md`。

主脚本树（与索引 §2 一致，快照）：

版本	脚本（均在 `scripts/explore/ripple_quantum_tests/`）	要点
v1	`ripple_quantum_tests.py`	NRMSE 阈值；默认 `artifacts/ripple_quantum_tests/`
v2	`ripple_quantum_tests_v2.py`	`shape_pass ∧ constant_pass`
v3	`ripple_quantum_tests_v3.py`	`derived` / `calibrated`；反作弊门
v4	`ripple_quantum_tests_v4_plot_optimize.py`	作图 + 差分进化；`artifacts/ripple_quantum_tests_v4/`
v5	`ripple_quantum_tests_v5_rigorous.py`	`nrmse_x` / `nrmse_y` / R^2；MRI κ；原子钟代数约束
v6	`ripple_quantum_tests_v6_joint.py`	联合 ((□,□,□,bw))；`--stress`；可选 **derived** 相速 + `ripple_medium_dispersion.py`
v8 合并	`ripple_quantum_tests_v8_unify.py`	**v8 unify** 主线（与 v7 图包脚本配套见下章）

36.3.1 一次运行快照（v1：闭式曲线，非优化）

v1 不做数值搜索，只在四条轴上比较 **QM-like 参考曲线** 与脚本内写定的 **Ripple 曲线** 的 **NRMSE**（定义见 `ripple_quantum_tests.py`）。下面是一次在仓库根目录、**默认阈值**下的可复跑记录（**2026-05-05**；`--seed 42` 对当前闭式实现几乎不改变数值，仅与输出元数据一致）。

项目	内容
命令	`python scripts/explore/ripple_quantum_tests/ripple_quantum_tests.py --threshold 0.18 --seed 42`
门禁	各子任务 `nrmse ≤ 0.18` 记为通过
产出	`artifacts/ripple_quantum_tests/RIPPLE_QUANTUM_TESTS_RESULTS.json`、…`_SUMMARY.md`、…`_PLOTS.png`（路径相对仓库根）

逻辑名	NRMSE（快照）	是否通过
`laser_threshold`	0.009503	是
`semiconductor_cutoff`	0.018966	是
`mri_larmor`	0.004509	是
`atomic_clock_modes`	0.072342	是

读法：**v1 全过**只说明「当前参数化下四条玩具形状贴得近」；**不**推出 SI 标定或替代量子力学。需要 **差分进化、双 NRMSE、R^2、MRI κ 代数、原子钟 L 约束** 时，改跑 **v5/v6**（一键命令见 `docs/RIPPLE_QUANTUM_TESTS_INDEX.md` §6）。

四条逻辑名（索引 §1）：`laser_threshold`、`semiconductor_cutoff`、`mri_larmor`、`atomic_clock_modes`——对应**玩具参考曲线**，**不是**公开实验 CSV。

本章边界
μ, ρ, η 等在本管线中多为**无量纲旋钮**；读者若把数值直接当物理常数用，需自备 **SI 映射**——本书此处**不装已解决。**

你多了一幅**「介质 toy 能否像四条参考曲线那样弯」**的地图。下一章收束：**热三元组预注册审计**、**v6 可辨识性**、**v7/v8 合并页**与**参数阱库**——把「专项审计」写成**可读的统一报告。**

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37 Chapter 12 •Ripple quantum four benchmarks v1–v8

涟漪量子四项玩具基准 v1–v8
Ripple quantum four benchmarks v1–v8

This chapter’s job
Introduce `scripts/explore/ripple_quantum_tests/`: on **four 1D axes**, use **dimensionless medium knobs** to match **QM-like reference shapes**; what versions **v1–v8** each add (gates, dispersion, merge pages, etc.).

37.1 12.1 Why this runs parallel to `ce_*`

`ce_*` is the **2D lattice + double slit / Bell** main front; the ripple-quantum line is **another** parameterized medium toy serving **four separate “industry curve”** intuitions: threshold, cutoff, Larmor linearity, line shape. It **does not** replace NIST event tables or claim **full SI calibration**.

37.2 12.2 What the four curves are

- **Laser**: power–pump shape near threshold.
- **Semiconductor**: features near the absorption edge.
- **MRI**: Larmor relation and algebraic **κ** (introduced in later versions).
- **Atomic clock**: spectrum and cavity-length algebra (see index for version notes).

37.3 12.3 Version ladder (v1→v8) and canonical index

Single navigation entry (in-repo): `docs/RIPPLE_QUANTUM_TESTS_INDEX.md` (bilingual, one-shot commands, `artifacts/` notes). Chinese task notes: `scripts/explore/ripple_quantum_tests/README.zh.md`.

Main script tree (matches index §2, snapshot):

Version	Script (all under `scripts/explore/ripple_quantum_tests/`)	Notes
v1	`ripple_quantum_tests.py`	NRMSE threshold; default `artifacts/ripple_quantum_tests/`
v2	`ripple_quantum_tests_v2.py`	`shape_pass ∧ constant_pass`
v3	`ripple_quantum_tests_v3.py`	`derived` / `calibrated`; anti-cheat gates

Version	Script (all under `scripts/explore/ripple_quantum_tests/`)	Notes
v4	`ripple_quantum_tests_v4_plot_optimize.py`	Plots + differential evolution; `artifacts/ripple_quantum_tests_v4/`
v5	`ripple_quantum_tests_v5_rigorous.py`	`nrmse_x` / `nrmse_y` / R^2; MRI κ; atomic-clock algebra
v6	`ripple_quantum_tests_v6_joint.py`	joint ((□,□,□,bw)); `--stress`; optional **derived** phase velocity + `ripple_medium_dispersion.py`
v8 merge	`ripple_quantum_tests_v8_unify.py`	**v8 unify** mainline (pairs with v7 figure bundle scripts —next chapter)

37.3.1 One-run snapshot (v1: closed form, no optimization)

v1 does no numerical search; it compares **NRMSE** between **QM-like reference curves** and script-defined **Ripple curves** on four axes (definition in `ripple_quantum_tests.py`). Rerunnable record from repo root, **default threshold**, **2026-05-05** (`--seed 42` aligns metadata; barely moves numbers for this closed-form build).

Item	Content
Command	`python scripts/explore/ripple_quantum_tests/ripple_quantum_tests.py --threshold 0.18 --seed 42`
Gate	Each subtask passes if `nrmse  0.18`
Outputs	`artifacts/ripple_quantum_tests/RIPPLE_QUANTUM_TESTS_RESULTS.json`, ... `_SUMMARY.md`, ...`_PLOTS.png` (paths relative to repo root)

Logical name	NRMSE (snapshot)	Pass
`laser_threshold`	0.009503	yes
`semiconductor_cutoff`	0.018966	yes
`mri_larmor`	0.004509	yes
`atomic_clock_modes`	0.072342	yes

How to read: **v1 all-pass** only means "under current parameterization the four toy shapes sit close" ; it does **not** imply SI calibration or replacing quantum mechanics. For **differential evolution, dual NRMSE, R², MRI κ algebra, atomic-clock L constraints**, run **v5/v6** (commands in `docs/RIPPLE_QUANTUM_TESTS_INDEX.md` §6).

Four logical names (index §1): `laser_threshold`, `semiconductor_cutoff`, `mri_larmor`, `atomic_clock_modes` —**toy reference curves**, **not** public experiment CSVs.

> **Chapter boundary**
> **μ, ρ, η** here are mostly **dimensionless knobs**; treating them as physical constants needs your own **SI map** —the book **does not pretend** that mapping is solved here.

You now have a map for **"can a medium toy bend like four reference curves?"** Next chapter tightens: **thermal-triplet preregistered audit**, **v6 identifiability**, **v7/v8 merge pages**, and the **parameter-trap library** —turning "focused audits" into a **readable unified report**.

38 第十三章•Chapter 13

涟漪专项审计与统一报告
Ripple-focused audits and unified report

> **本章任务**
> 把第十二章散落的**硬门禁**收成**叙事**：热波三元组盲测、全频透射、三体恒等式、可辨识性崩溃案例、合并 HTML 报告——**哪些是成果，哪些是反例库。**

38.1 13.1 热三元组与预注册

- `scripts/explore/ripple_quantum_tests/ripple_thermal_triplet_preregistered_audit.py`
  预注册门 + 盲测/反作弊叙事；与第三章 **PDE toy** 同态度：**规则先写死，再谈通过/失败。**

38.2 13.2 全频、三体恒等式与可辨识性

下列除单独注明外，均在 `scripts/explore/ripple_quantum_tests/`：

脚本	大致用途
`ripple_triplet_fullband_transparency_audit.py`	**全频透射**审计
`ripple_triplet_physics_identity_check.py`	**三体物理恒等式**核对
`ripple_v6_identifiability_audit.py`	**v6 可辨识性**（平坦/多谷 → 明确失败标签）
`rho_mu_feasibility_map.py`、`rho_sensitivity_breakdown.py`、`verify_rho_mu2_constraint.py`	**ρ–μ 可行域**与灵敏度分解
`v10_fail_anatomy.py`	**失败解剖**（为何不过门）

参数阱：多脚本交叉扫出来的「好看但不稳」角落——本书视为**反例库**，不是污点。

38.3 13.3 v7/v8 图包与合并 HTML

- `generate_v7_paper_figures.py`、`generate_v8_paper_figures.py`：论文级图包导出。
- `build_merged_v7_v8_html.py`：**v7 / v8 合并页**，多曲线一屏对照。

读法：**先看图例与坐标定义**，再比**参考 vs 涟漪侧**与**范数定义。**

> **本章边界**
> 「统一报告」是**仓库内**归档组织方式；**不**等于期刊论文已接收。引用图表请**连带脚本 commit 或 artifact hash**（若归档内有）。

涟漪线至此从**玩具**走到**有门禁的审计链**。下一章**兜底**：Born、不确定性、退相干等**其他 explore 基线**，并指向 `explore_*` **索引**互见。

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39 Chapter 13 •Ripple-focused audits and unified report

涟漪专项审计与统一报告
Ripple-focused audits and unified report

> **This chapter’s job**
> Turn Chapter 12’s scattered **hard gates** into narrative: thermal-wave triplet blind test, full-band transmission, triplet physics identity, identifiability-failure cases, merged HTML reports —**what counts as a win vs a counterexample library**.

39.1 13.1 Thermal triplet and preregistration

- `scripts/explore/ripple_quantum_tests/ripple_thermal_triplet_preregistered_audit.py`
  Preregistered gates + blind-test / anti-cheat narrative; same attitude as Chapter 3’s **PDE toy**: **freeze rules first, then talk pass/fail.**

39.2 13.2 Full band, triplet physics identity, identifiability

Unless noted, all under `scripts/explore/ripple_quantum_tests/`:

Script	Rough role
`ripple_triplet_fullband_transparency_audit.py`	**Full-band transmission** audit
`ripple_triplet_physics_identity_check.py`	**Triplet physics identity** check
`ripple_v6_identifiability_audit.py`	**v6 identifiability** (flat / multi-valley → explicit failure labels)
`rho_mu_feasibility_map.py`, `rho_sensitivity_breakdown.py`, `verify_rho_mu2_constraint.py`	**ρ–μ feasible sets** and sensitivity breakdown
`v10_fail_anatomy.py`	**Failure anatomy** (why a gate failed)

Parameter traps: corners that look good but are unstable under cross-scans —treated here as a **counterexample library**, not a stain.

39.3 13.3 v7/v8 figure bundles and merged HTML

- `generate_v7_paper_figures.py`, `generate_v8_paper_figures.py`: publication-style figure export.
- `build_merged_v7_v8_html.py`: **merged v7/v8 page**, many curves on one screen.

Reading order: **legend and axis definitions first**, then compare **reference vs ripple side** and **norm definitions**.

> **Chapter boundary**
> “Unified report” is **in-repo** packaging; it is **not** “accepted at a journal.” Cite figures with **script commit or artifact hash** when available.

The ripple thread now moves from **toys** to a **gated audit chain**. Next chapter is the **catch-all**: Born, uncertainty, decoherence, and other `explore_*` **baselines**, with pointers to the `explore_*` **index**.

40 第十四章•Chapter 14

其他基线与探索
Other baselines and exploration

> **本章任务**
> 说明仓库中**未编入前述主线、仍值得知道**的探索：`verify_*`、`discover_*`、部分 `explore_critique_*`——**Born 规则抽查**、**不确定性扫描**、**退相干 toy**、**相对论格子批评**等。

40.1 14.1 为什么需要「杂物抽屉」章

若只读 `ce_*` 与 NIST，会误以为全书**只有双缝与 Bell**。实际上，仓库长期用**小脚本**回答：**某条经典结论在格点上是否稳定复现**、**哪一步推广会踩雷**。本章**不逐文件讲**，只给**地图**。

40.2 14.2 `scripts/verify/`（一次性基线）

下列路径均为 `scripts/verify/`：

文件	大致用途
`verify_born_rule.py`	**Born 规则**式概率在格点设定下是否自洽
`verify_delayed_choice.py`	**延迟选择**相关基线（与 `ce_06` 叙事互证）
`verify_interference_decay.py`	**干涉衰减**行为
`verify_uncertainty.py`	**不确定性**关系 toy
`verify_which_way.py`	**which-way** 类对照

40.2.1 可复跑快照：`verify_born_rule.py`

连续场屏分布 vs 蒙特卡洛光子累计（`ce_engine_v2`）。默认 **N_PHOTONS=50000**、**seed=42**；运行约**数十秒** 量级（视 CPU）。**图**已固定写入 `scripts/verify/verify_born_rule.png`（与脚本同目录，不依赖 cwd）。

项目	内容
命令	在仓库根：`PYTHONPATH` 指向仓库根后执行 `python scripts/verify/verify_born_rule.py`
快照日期	**2026-05-05**（默认常量，未改 `N_PHOTONS`）

`observed` 键	值（快照）
`pearson_r_continuous_vs_mc`	≈ **0.9441**
`visibility_continuous`	≈ **0.6859**
`visibility_mc`	≈ **0.1646**
`hit_rate`	≈ **0.0421**（击中屏光子占比）

读法：**皮尔逊 r** 衡量两列**形状对齐**，与 `ce_05` 里那种单列 **V** 不是同一对象；MC 的 **V** 低而 **r** 仍可高，因计数噪声与归一化方式不同。**档案假设**里已写明：**r 高不代表在测度论意义上证明了 Born**——本书把该脚本当作 **「同一引擎下两场叙事是否同形」**的基线，不把脚本末尾的修辞句当判决书。

40.2.2 可复跑快照：`verify_delayed_choice.py`

`ce_engine_v2` 上先长跑再对场做**行域裁剪/放大**（模拟「过缝后再选路径」），与 `ce_06`「第几步插入圆吸收器」是**同一类——因果顺序写进循环**，但**实现细节不同**。图写入 `scripts/verify/verify_delayed_choice.png`。

项目	内容
命令	在仓库根：`PYTHONPATH` 指向仓库根后执行 `python scripts/verify/verify_delayed_choice.py`
快照日期	**2026-05-05**（默认常量）

`observed` 键	值（快照）
`vis_no_measurement`	≈ **0.5613**
`vis_delayed_slit_mask`	**0.0**
`verdict_strict_qm_style`	**true**（脚本内置：`V_` 无测量 `>0.3` 且 `V_` 延迟掩模 `<0.15`）
`verdict_trend_only`	**false**

读法：这里的「延迟测量」是**网格上的显式掩模代数**，不是实验室光子学；**V=0** 来自 `compute_visibility` 在裁剪后**找不到有效峰谷**。与 `ce_06` 对照阅读：圆斑吸收 vs 行带清零，何者在几何上更「钝」、何者更「狠」，可以用**同一屏列定义**做表。勿把脚本里的口号式 stdout 当全书结论——**档案** `reviewer_prompts` 已邀请「经典场论类比是否完全等价」类质疑。

40.3 14.3 `scripts/discover/`（现象扫描）

下列路径均为 `scripts/discover/`：

文件	大致用途
`discover_visibility_decay.py`	**可见度**随参数衰减
`discover_coupling_constant.py`	**耦合常数**敏感区
`discover_measurement_continuity.py`	**测量/连续**参数扫描

40.4 14.4 `explore_critique_*`（内部批评）

均在 `scripts/explore/`：

文件	主题（见脚本内标题）
`explore_critique_01_unification_scope.py`	统一范围
`explore_critique_02_bell_hypothesis_boundary.py`	Bell 假设边界
`explore_critique_03_analogy_vs_mechanism.py`	类比 vs 机制
`explore_critique_04_sr_not_derived_from_lattice.py`	相对论未从格子导出
`explore_critique_05_decay_nonuniqueness.py`	衰减非唯一
`explore_critique_06_energy_growth_explosion.py`	能量增长
`explore_critique_07_fringe_spacing_theory_gap.py`	条纹间距理论缝

与序言 **「非法典」**一致：**类比可以养直觉，机制要另过门。**

40.4.1 可复跑示例：`explore_critique_03`（措辞门禁，无数值物理）

`scripts/explore/explore_critique_03_analogy_vs_mechanism.py` 不跑格子，只打印三条**对外表述建议**并生成示意图，stdout 带 `[OK] critique_03_analogy_language`。适合当作「写论文/写书前过一遍」的自动化便签。

项目	内容
命令	`python scripts/explore/explore_critique_03_analogy_vs_mechanism.py`
产出图	`scripts/explore/explore_critique_03_analogy_vs_mechanism.png`
`observed` **标记**	`marker: [OK] critique_03_analogy_language`（**2026-05-05** 跑通快照）

40.5 14.5 如何自己索引

涟漪子树以 `docs/RIPPLE_QUANTUM_TESTS_INDEX.md` 为纲；其余以 `scripts/explore/` 目录 + 文件名为纲。**命名即意图**：`explore_` 多扫描，`verify_` 多一次性检验，`discover_` 多现象发现。

> **本章边界**
> 「探索」**不**都是成稿结论；部分脚本保留**失败输出**。引用前请**跑一遍**或读 artifact 注释。

仓库的**可复现性**不仅来自主线，也来自这些**小基线**与**自我批评脚本**。下一章为**规划占位**：**Bell 类公开数据上 Alice/Bob 球面坐标联合拟合**（**文稿先行，代码待补**）；再下一章才是**诚实列出没通过的测量**。

路标：**栅格 + NIST + CHSH 审计**的主线在中腹已大体铺完；**第 12–13 章的涟漪量子 toy** 是**并行探索**，曲线再漂亮也要先问门禁与失败记录。若你**只要公开数据与读数规则的诚实记录**，可侧重**第 8–10 章与第 16 章**；**第 17–18 章**为**设想与动力系统草图**，读来完全可选。

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41 Chapter 14 •Other baselines and exploration

其他基线与探索
Other baselines and exploration

> **This chapter’s job**
> Map explorations **not wired into the main line but still worth knowing**: `verify_*`, `discover_*`, selected `explore_critique_*` —**Born spot checks**, **uncertainty scans**, **decoherence toys**, **relativistic-lattice critique**, etc.

41.1 14.1 Why a “junk drawer” chapter

Reading only `ce_*` and NIST can imply the repo is **only double slit and Bell**. In practice, **small scripts** ask whether **classic conclusions reproduce stably on the grid** and **which generalization blows up**. This chapter **does not walk file-by-file**; it gives a **map**.

41.2 14.2 `scripts/verify/` (one-shot baselines)

All paths `scripts/verify/`:

File	Rough role
`verify_born_rule.py`	Whether **Born-style** probabilities stay coherent on this lattice setup
`verify_delayed_choice.py`	Delayed-choice baseline (cross-read `ce_06`)
`verify_interference_decay.py`	**Interference decay** behavior
`verify_uncertainty.py`	**Uncertainty** relation toy
`verify_which_way.py`	**Which-way**-style controls

41.2.1 Rerunnable snapshot: `verify_born_rule.py`

Continuous field screen distribution vs Monte Carlo photon accumulation (`ce_engine_v2`). Defaults **N_PHOTONS=50000**, **seed=42**; run time **tens of seconds** (CPU dependent). Figure pinned to `scripts/verify/verify_born_rule.png` (same directory as script, cwd-independent).

Item	Content
Command	From repo root with `PYTHONPATH`: `python scripts/verify/verify_born_rule.py`
Snapshot date	**2026-05-05** (default constants, `N_PHOTONS` unchanged)

`observed` key	Value (snapshot)
`pearson_r_continuous_vs_mc`	≈ **0.9441**
`visibility_continuous`	≈ **0.6859**
`visibility_mc`	≈ **0.1646**
`hit_rate`	≈ **0.0421** (fraction of photons hitting screen)

How to read: **Pearson r** measures **shape alignment between two columns** —not the same object as single-column **V** in `ce_05`; MC **V** can be low while **r** stays high because of counting noise and normalization. Assumptions in the dossier state clearly: **high r is not a measure-theoretic proof of Born** —the script is a baseline for **"do two narratives on the same engine share shape?"** ; do not treat closing rhetoric in stdout as a verdict.

41.2.2 Rerunnable snapshot: `verify_delayed_choice.py`

On `ce_engine_v2`, long run then **row-domain crop / zoom** on the field ("choose path after the slits"), same **family** as `ce_06`' s "which step inserts the round absorber" —**causal order in the loop**, **different implementation**. Figure: `scripts/verify/verify_delayed_choice.png`.

Item	Content
Command	From repo root with `PYTHONPATH`: `python scripts/verify/verify_delayed_choice.py`
Snapshot date	**2026-05-05** (defaults)

`observed` key	Value (snapshot)
`vis_no_measurement`	≈ **0.5613**
`vis_delayed_slit_mask`	**0.0**

observed key	Value (snapshot)
verdict_strict_qm_style	**true** (built-in: V_no_meas > 0.3 and V_delayed_mask < 0.15)
verdict_trend_only	**false**

How to read: "Delayed measurement" here is **explicit mask algebra on the grid**, not lab photonics; **V = 0** because compute_visibility finds **no usable peaks/troughs** after cropping. Cross-read ce_06: round absorber vs row zeroing —which geometry is "blunter" vs "harsher" —tabulate under **the same screen-column definition**. Do not treat slogan stdout as the book's conclusion —dossier reviewer_prompts already invites "is the classical-field analogy fully equivalent?" style doubt.

41.3 14.3 scripts/discover/ (phenomenon scans)

All scripts/discover/:

File	Rough role
discover_visibility_decay.py	**Visibility** vs parameter decay
discover_coupling_constant.py	**Coupling** sensitive regions
discover_measurement_continuity.py	**Measurement / continuity** scans

41.4 14.4 explore_critique_* (internal critique)

All scripts/explore/:

File	Theme (see in-file title)
explore_critique_01_unification_scope.py	Unification scope
explore_critique_02_bell_hypothesis_boundary.py	Bell hypothesis boundary
explore_critique_03_analogy_vs_mechanism.py	Analogy vs mechanism
explore_critique_04_sr_not_derived_from_lattice.py	SR not derived from lattice
explore_critique_05_decay_nonuniqueness.py	Decay non-uniqueness
explore_critique_06_energy_growth_explosion.py	Energy growth
explore_critique_07_fringe_spacing_theory_gap.py	Fringe-spacing theory gap

Matches preface **"not a codex"**: **analogies may train intuition; mechanisms must pass their own gates.**

41.4.1 Rerunnable example: explore_critique_03 (language gate, no numerical physics)

scripts/ explore/ explore_critique_03_analogy_vs_mechanism.py does not run the lattice; it prints three **outward phrasing suggestions** and emits a schematic; stdout carries [OK] critique_03_analogy_language. Handy automated sticky note **before drafting a paper or chapter**.

Item	Content
Command	python scripts/explore/ explore_critique_03_analogy_vs_mechanism.py
Figure	scripts/explore/ explore_critique_03_analogy_vs_mechanism.png

Item	Content
`observed` **marker**	`marker: [OK] critique_03_analogy_language` (**2026-05-05** run)

41.5 14.5 How to index yourself

Ripple subtree: `docs/RIPPLE_QUANTUM_TESTS_INDEX.md`; otherwise `scripts/explore/` by directory + filename. **Names signal intent**: `explore_` scans, `verify_` one-off checks, `discover_` phenomenon sweeps.

> **Chapter boundary**
> "Exploration" is **not** always a finished conclusion; some scripts keep **failure outputs**. **Rerun** or read artifact notes before citing.

Reproducibility here comes from the **main line** and from these **small baselines** plus **self-critique scripts**. Next is a **planning placeholder**: **joint Alice/Bob sphere fit on Bell-type public data** (**manuscript first, code TBD**); after that comes the **honest list of measurements that did not pass**.

Layer cue: The **grid + NIST + CHSH audit spine** is largely set by mid-book; **Chapters 12–13 ripple-quantum toys** are **parallel exploration** —ask which gates a curve passed before admiring it. If you mainly want **public-data / readout honesty**, lean on **Chapters 8–10 and 16**; **Chapters 17–18** are **optional sketches** of ideas and dynamics.

42 第十五章•Chapter 15

(规划稿）Bell 类公开数据：联合拟合 Alice 与 Bob 的球面坐标
Planned: Bell-type public data —joint fit of Alice's and Bob's sphere coordinates

> **本章任务（当前为占位）**
> 预留**文稿位置**：说明**拟做什么、与第 9 章及第 8–10 章审计链如何衔接**；**正文实现（脚本、JSON、图）待补。**

42.1 15.1 为什么要单开一章（而不是塞进第 9 章）

- **第 9 章**已经承担「**NIST 数据 → 庞加莱球等效坐标**」的**独立拟合**叙事；读者应**先读该章**建立语言。
- **Bell/CHSH 子问题**常额外要求：**哪些 trial 进入 S、二值化阈值是否与拟合假设打架**——若在**同一管道**里混写，容易**口头合并、账目分开。**本章预留为 **「Bell 聚焦 + 联合 Alice/Bob」**的**专用抽屉。**
- **诚实边界**：**未实现**不等于**不重要**；先**把意图写死**，避免日后草稿散在 issue 里找不到。

42.2 15.2 计划中的技术轮廓（无代码承诺）

下列条目**描述意图**，**不构成**已实现接口：

块	计划内容
输入	与 `battle_results/` 或 `papers_final/` 中已冻结的 Bell/CHSH 事件表对齐的 CSV/HDF5；字段语义遵循第 8–10 章。
输出	每侧（Alice / Bob）**球面坐标轨迹**或**分段常值** + **不确定度**；可选：**与 CHSH 四设定同步索引。**

块	计划内容
对照	与 **第 9 章**拟合结果**差分**（同一数据子集）；**阴性对照**：打乱 trial 顺序、伪造标签。
门禁	**可辨识性**崩溃则**不写胜利句**；**metrics.json + verdict.md** 式归档（与全书习惯一致）。

42.3 15.3 作者说明：为何此刻只有纸面

实现这一章需要**连续投入**：数据对齐、优化器、诊断图、与现有 NIST 管道**回归测试。**作者当前**动力不足**，选择**先把章节占住、把边界写清**，以免日后补上时与全书**体例冲突。**

> **本章边界**
> **规划稿**；**无**可运行脚本、**无**新实验数据。读者**不得**把本章当作已完成工作引用。下一章回到**已写内容**：**没通过的测量**清单。

球面上的 Alice/Bob 是**给仪器和统计对齐用的语言**；本书预留一章把 **Bell 类公开数据** 接进这套语言——**待办。**读完请直接去 **第十六章**：那里记录**已经发生的失败与未闭合。**

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43 Chapter 15 •Planned: Bell-type public data —joint fit of Alice’s and Bob’s sphere coordinates

(规划稿）Bell 类公开数据：联合拟合 Alice 与 Bob 的球面坐标
Planned: Bell-type public data —joint fit of Alice’s and Bob’s sphere coordinates

> **This chapter’s job (placeholder)**
> Reserve **manuscript space**: what is intended, how it hooks **Chapter 9** and the **Chapters 8–10 audit chain**; **implementation (scripts, JSON, figures) TBD**.

43.1 15.1 Why a separate chapter (not folded into Chapter 9)

- Chapter **9** already owns **NIST data → independent Poincaré fit**; read it first for language.
- **Bell/CHSH subproblems** often need **which trials enter S, whether binarization thresholds fight fit assumptions** —mixing into one pipeline risks **merged rhetoric, split ledgers**. This slot is the **“Bell-focused + joint Alice/Bob”** drawer.
- **Honest boundary**: **unimplemented ≠ unimportant**; **nail intent in text** so drafts do not scatter across issues.

43.2 15.2 Planned technical outline (no code promise)

These lines **describe intent**, **not** shipped interfaces:

Block	Planned content
Input	CSV/HDF5 aligned to frozen Bell/CHSH event tables under `battle_results/` or `papers_final/`; field semantics follow Chapters 8–10.

Block	Planned content
Output	Per side (Alice / Bob) **sphere-coordinate traces** or **piecewise constants** + **uncertainty**; optional: **index locked to CHSH’s four settings**.
Controls	**Diff** vs Chapter **9** fits on the **same subset**; **negative controls**: shuffle trials, fake labels.
Gates	If **identifiability** collapses, **no victory sentences**; archive `metrics.json + verdict.md` style like the rest of the book.

43.3 15.3 Author note: why only paper for now

Shipping this chapter needs **sustained work**: data alignment, optimizers, diagnostic plots, **regression tests** against existing NIST plumbing. The author is **low on energy** right now and chose to **hold the chapter slot and write boundaries** so a future implementation does not **fight house style**.

> **Chapter boundary**
> **Planning draft**; **no** runnable scripts, **no** new experimental data. **Do not** cite this chapter as finished work. Next chapter returns to **written material**: a **list of measurements that did not pass**.

Alice/Bob on the sphere is **language for aligning instruments and statistics**; this book reserves a chapter to plug **Bell-type public data** into that language —**todo**. Go next to **Chapter 16**: **failures and open loops already on record**.

44 第十六章•Chapter 16

没通过的测量：诚实记录
What failed: an honest record

> **本章任务**
> 把「失败」从情绪变成**清单**：哪些假设在数据或 in-silico 上**未闭合**、哪些曲线**未过门**、哪些**叙事曾写早**——并说明**是否归档、如何复查。**

44.1 16.1 失败为什么要入册

若一本书只写胜利，读者无法知道**边界在哪。**本书把**未通过的门禁**、**未复现的对照**、**可辨识性崩溃**与**参数阱**当作**一等公民**：它们告诉你：**别把钱押在尚未钉死的推广上。**

44.2 16.2 几类「没通过」

- **模型内**：某次 `ce_*` / `ghz_*` / `ripple_*` 扫描**未达阈值**或**与阴性对照冲突。**
- **审计层**：某条 NIST 管道在**改分母**后结论**不再稳定**；或**子样本**不足以下判断。
- **解释层**：某段口语推广被 `explore_critique_*` 内部否决——**类比保留，机制不升格。**

44.3 16.3 如何复查（脚本与目录线索）

目录：`battle_results/`、`artifacts/`、`papers_final/` ——优先文件名含 `fail`、`red`、`closure`、`anatomy`、`honest`、`hardcore` 的报告或 JSON。

脚本级「失败解剖」示例（非穷尽）：

- `scripts/explore/ripple_quantum_tests/v10_fail_anatomy.py`
- `scripts/explore/ripple_quantum_tests/ripple_v6_identifiability_audit.py`
- `scripts/explore/ghz_honest_cost_curve_v11.py`、`ghz_hardcore_audit_v18.py`（命名即态度）
- `analyze_v3_open_loops.py`（开放环/未闭合诊断）

复查纪律：**同一 commit** 或 **artifact 内嵌 hash** 对齐版本；**禁止**只贴一张截图不附脚本与数据版本。

> **本章边界**
> 「没通过」**不是**对任何实验「判死刑」；它是**本书当前版本**在**自述规则**下的状态。下一版可能推翻——前提是**规则与数据一并更新**。

假大象要量，也要**量不出来的地方**。第三部分换节奏：**设想**两条——**经典光力学计算路径**与**涟漪动力系统**——明确标为**构想**，不是成品。

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45 Chapter 16 •What failed: an honest record

没通过的测量：诚实记录
What failed: an honest record

> **This chapter’s job**
> Turn “failure” into a **list**: which assumptions stayed **open** on data or in-silico, which curves **missed gates**, which narratives were **premature** —and **whether archived, how to re-audit**.

45.1 16.1 Why failures belong in the book

If a book only reports wins, readers cannot see **where the edge is**. This book treats **failed gates**, **unreproduced controls**, **identifiability collapses**, and **parameter traps** as **first-class citizens**: they say **do not bet on promotions that are not yet nailed down**.

45.2 16.2 Kinds of “did not pass”

- **In-model**: a `ce_*` / `ghz_*` / `ripple_*` scan **missed thresholds** or **conflicts with negative controls**.
- **Audit layer**: a NIST pipeline **stops being stable** after **denominator edits**; or **subsamples** too small to judge.
- **Explanation layer**: colloquial hype vetoed by `explore_critique_*` —**keep analogy, do not upgrade mechanism**.

45.3 16.3 How to re-audit (directories and script hints)

Directories: `battle_results/`, `artifacts/`, `papers_final/` —prefer filenames with `fail`, `red`, `closure`, `anatomy`, `honest`, `hardcore`.

Script-level “failure anatomy” examples (non-exhaustive):

- `scripts/explore/ripple_quantum_tests/v10_fail_anatomy.py`
- `scripts/explore/ripple_quantum_tests/ripple_v6_identifiability_audit.py`
- `scripts/explore/ghz_honest_cost_curve_v11.py`, `ghz_hardcore_audit_v18.py` (names signal intent)
- `analyze_v3_open_loops.py` (open-loop / non-closure diagnostics)

Re-audit discipline: align **same commit** or **artifact-embedded hash**; **do not** post a lone screenshot without script + data version.

> **Chapter boundary**
> "Did not pass" is **not** a death sentence on any experiment; it is **this book's current build** under **its own stated rules**. A later edition may reverse —only if **rules and data update together**.

The **fake elephant** must be measured —and **where measurement stops**. Part III shifts tempo: two **sketches** —**classical optical computing pathway** and **ripple propulsion** —explicitly labeled **speculation**, not product.

46 第十七章•Chapter 17

经典光力学计算路径的设想
A classical optical computing pathway

> **本章任务**
> **构想章**：若坚持**局域传播 + 可调损耗 + 显式测量规则**，能否在工程上走出一条**经典光学式**的信息处理路径——**不**宣称已替代量子器件，只列**必要条件**与**已知的硬天花板**。

46.1 17.1 从格子到器件想象

前几章的格子是**思想与审计的试验田**。设想把它**芯片化**：波导、分束、可调吸收、相位片——每一层都对应**参数化规则**。优势是**透明**；风险是**尺度与噪声**把「干净定理」磨糊。

46.2 17.2 仓库里已有的「算力」隐喻（不是成品光芯片）

本书**尚未**实现商用光计算流片；但有两条**可对照的仓库线索**，说明「经典路径」在**方法论**上不是空话：

线索	路径	与「光力学计算」的关系
网格 + 协议	`scripts/ce/*`、`chain_explosion_numba.py`	**可编程规则**已在 Python 里跑通：传播、吸收、分支、读出——相当于**软件定义的光学积木**（数字 toy）。
最小计算 ISA 探索	**计算机探索**/`README_` **探索路线**`.md` 及同目录脚本	在**禁止 sqrt / 牛顿迭代**等约束下，讨论「介质动力学能否完成可重复映射」——是**经典计算哲学**的极端洁癖版，与工程光计算**同族不同阶**。
研究地图	`docs/RESEARCH_MAP.md` 第三层「计算与审计」	强调 **可复现脚本替代口头争论**；光计算若要走通，最终也要过**同一类门禁**。

读法：**先把** `ce_*` **当「逻辑门级仿真」理解**，再谈是否映射到 **EDA/FDTD**；中间缺的那几层，正是本章所说的**里程碑**。

46.3 17.3 与量子路线的关系

本书**不**参与「量子霸权」口水战。务实问题是：**哪类任务**在**误差与能耗约束**下仍值得用经典波动模拟；**哪类任务**公开证据已显示**优势在量子侧**。`RESEARCH_MAP.md` 第二层把 **Bell/CHSH 统计对象**定为「核心战场」——经典光路**不**自动退出战场，但**多体相关**仍受 `07-archive-ghz.md` 一类论证约束。

46.4 17.4 需要的里程碑（工程化）

- **协议语言**与工业 **EDA / 光电仿真**对齐（不仅是 Python toy）。
- **可制造性**：**λ、η、几何**的批次波动是否纳入**良率模型**。
- **审计链**：从 **layout → 仿真 → 流片测试** 能否保留本书坚持的 **JSON/版本/hash** 习惯（见专论与 `experiment_dossier` 思想）。

本章边界
全章为**设想**；**无**新实验数据。若未来落地，应**另开技术专册**，本书不升格为工程手册。

经典路径不是退路，是**一条需要重新定价的路**：仓库已证明**规则可编码、可审计**；**未证明**的是**尺度、噪声与多体边界**下的性价比。下一章换更「宏观」的涟漪想象：**动量/能量账本**——同样**只作构想**，但会挂钩 `unsolved_problems_collision` 里的**可证伪指标**思路。

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47 Chapter 17 •A classical optical computing pathway

经典光力学计算路径的设想
A classical optical computing pathway

This chapter’s job
Sketch: With **local propagation + tunable loss + explicit measurement rules**, could an engineering path emerge for **classical optics–style** information processing —**without** claiming to replace quantum hardware; list **necessary conditions** and **known hard ceilings**.

47.1 17.1 From lattice to device imagination

Earlier chapters’ lattice is a **testbed for ideas and audits**. Imagine **chip-scale** realization: waveguides, splitters, tunable absorbers, phase plates —each maps to **parameterized rules**. Upside: **transparency**; risk: **scale and noise** blur “clean theorems.”

47.2 17.2 “Compute” metaphors already in the repo (not a commercial photonic chip)

No commercial optical-compute tape-out yet; two **repo clues** show the classical path is **not hollow methodologically**:

Thread	Path	Relation to "optical computing"
Grid + protocol	`scripts/ce/*`, `chain_explosion_numba.py`	**Programmable rules** already run in Python: propagate, absorb, branch, read out — **software-defined optical blocks** (digital toy).
Minimal compute ISA exploration	**计算机探索**`/README_`**探索路线**`.md` and sibling scripts	Under bans like **no sqrt / Newton iteration**, asks whether **medium dynamics can implement repeatable maps** —an extreme **classical computing philosophy** exercise, **same family, different rung** than engineering photonics.
Research map	`docs/RESEARCH_MAP.md`, layer three "compute and audit"	Stresses **reproducible scripts instead of oral argument**; optical computing, if serious, must pass **the same style of gates**.

Reading path: treat `ce_*` **as "logic-level simulation"** first, then ask whether it maps to **EDA/FDTD**; the missing middle layers are the **milestones** this chapter names.

47.3 17.3 Relation to quantum routes

This book **does not** join "quantum supremacy" shouting matches. Practical question: **which tasks** still reward classical wave simulation under **error and energy budgets**; **which tasks** public evidence already parks on the **quantum side**. `RESEARCH_MAP.md` layer two pins **Bell/CHSH statistical objects** as a "core battlefield" —classical optics **does not** automatically leave the field, but **many-body correlation** still faces arguments like `07-archive-ghz.md`.

47.4 17.4 Milestones if you industrialize

- **Protocol language** aligned with **EDA / optoelectronic simulation** (not only Python toys).
- **Manufacturability**: batch spread of **λ, η, geometry** inside a **yield model**.
- **Audit chain**: can **layout → simulation → fab test** keep this book's **JSON / version / hash** habits (see monograph and `experiment_dossier` spirit).

  Chapter boundary
  Whole chapter is **speculation**; **no** new experimental data. If it ever ships, it deserves a **technical companion volume** —this book does not upgrade to an engineering handbook.

The **classical path** is not a retreat; it is a **route that needs repricing**. The repo already shows **rules can be encoded and audited**; what is **not** proved is **cost–benefit under scale, noise, and many-body edges.** Next chapter widens the aperture: **momentum / energy ledgers** for a **ripple propulsion** sketch —still **speculation**, but hooks `unsolved_problems_collision` **falsifiable-metrics** style.

48 第十八章•Chapter 18

涟漪动力系统的设想
A ripple-based propulsion concept

> **本章任务**
> **构想章**：把**介质里传播的扰动**想象成一种**可记账的动量/能量交换**——讨论**在何种对称性与损耗模型下**，「涟漪推进」在**原理上**可能说什么、**不可能**说什么。

48.1 18.1 从科普隐喻到方程纪律

「涟漪推船」在口语里好听，在物理里要先问：**动量去哪了、介质反冲算不算、开放系统熵增在哪。**本书前几章的格子**没有**自动包含**宏观 Navier–Stokes**；因此本章**不**给出具体推力公式，只列**若要认真，必须补哪类连续极限。**

仓库内最接近「动量/能量口径」的可操作讨论，见 `unsolved_problems_collision/ROADMAP_7_PROBLEMS.md` **第 1 题**（亚伯拉罕–闵可夫斯基动量表述之争的 toy 化）：文中给出 **前沿速度比、能流积分、界面反冲代理**等**可证伪指标**与**失败判据**——这正是把「涟漪推进」从**修辞**拉回**指标**的示范格式。**本书不宣称该路线图已闭合**；只建议：**谁若谈推进，谁先认领一套类似 `I1/I2/I3` 的数。**

48.2 18.2 与第三章「热」的衔接

若有**净推力**，往往伴随**不对称损耗**或**不对称辐射**——第三章里「漏进浴」的份额，在这里可能变成**定向动量。**任何具体数值都需要把 **bath 自由度**或**辐射场**写清，否则只是**词语游戏。**

48.3 18.3 与「相对论未从格子导出」的提醒

`scripts/explore/explore_critique_04_sr_not_derived_from_lattice.py` 内部立场：当前离散格子**没有**自动推出**洛伦兹对称**——若「推进」叙事偷偷借用相对论动量公式，却未建立**模型与 SR 的接口**，属于**范畴错配。**推进构想必须先说清：**你在哪一套极限/哪一类对称性下谈动量。**

48.4 18.4 开放问题

- 离散格子**能否**、**在何尺度下**与某类**连续介质方程**严格对接（见 `00-unified-writing-scheme.md` 与 `01-appendix-three-phrasings.md`）。
- **可证伪预测**：沿用 `ROADMAP_7_PROBLEMS.md` 的写法，为「宏观推进」锁定 **metrics.json + verdict.md** 式输出——**本书尚未**代你做完这一步。

> **本章边界**
> **纯设想**；**禁止**把本章当专利说明书或工程可行性背书。

泡泡可以讲故事；**推进**要写**方程与账本**——仓库里已有**路线图模板**（`ROADMAP_7_PROBLEMS.md`），缺的只是你是否愿意把口号压成**指标与失败判据。**后记收束全书。

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49 Chapter 18 •A ripple-based propulsion concept

涟漪动力系统的设想
A ripple-based propulsion concept

This chapter's job
Sketch: imagine **disturbance propagating in a medium** as **bookable momentum / energy exchange** —under **which symmetry and loss models** "ripple propulsion" can **say** something in principle and **cannot**.

49.1 18.1 From pop metaphor to equation discipline

"Ripple pushes the boat" sounds good in prose; in physics ask first: **where momentum went, whether medium recoil counts, where open-system entropy production lives.** Earlier lattice chapters **do not** automatically embed **macro Navier–Stokes**; this chapter **does not** give a thrust formula —only **which continuum limit you must add** to be serious.

Closest in-repo, operational momentum/ energy discussion: `unsolved_problems_collision/ROADMAP_7_PROBLEMS.md` **problem 1** (Abraham–Minkowski toy): **front-velocity ratios, energy-flux integrals, interface-recoil proxies** as **falsifiable metrics** and **failure criteria** —a template for dragging "ripple propulsion" from **rhetoric** to **numbers**. **This book does not claim that roadmap is closed**; it only suggests: **whoever talks thrust should own numbers in the I1/I2/I3 spirit.**

49.2 18.2 Bridge to Chapter 3 "heat"

Net thrust often rides with **asymmetric loss** or **asymmetric radiation** —the share that Chapter 3 "leaks into the bath" can become **directed momentum** here. Any concrete number needs **bath degrees of freedom** or a **radiation field** written explicitly; otherwise it is **wordplay**.

49.3 18.3 Reminder: "SR not derived from lattice"

`scripts/explore/explore_critique_04_sr_not_derived_from_lattice.py` states: the current discrete grid **does not** automatically yield **Lorentz symmetry** —if a "propulsion" story quietly borrows relativistic momentum formulas without a **model–SR interface**, that is a **category mistake**. Any propulsion sketch must first say **which limit and which symmetry class** define "momentum."

49.4 18.4 Open questions

- Can the discrete lattice **lock**, and **at what scale**, to a chosen **continuum medium equation** (see `00-unified-writing-scheme.md` and `01-appendix-three-phrasings.md`).
- **Falsifiable prediction**: reuse `ROADMAP_7_PROBLEMS.md` style to pin **macro propulsion** to `metrics.json + verdict.md` outputs —**this book does not** do that step for you.

Chapter boundary
Pure speculation; **do not** treat this chapter as a patent filing or engineering sign-off.

Bubbles can tell stories; **propulsion** needs **equations and ledgers** —the repo already has a **roadmap template** (`ROADMAP_7_PROBLEMS.md`); what is missing is willingness to compress slogans into **metrics and failure criteria.** The afterword closes the book.

50 后记

你刚读完的，是一本用**泡泡直觉**搭**假大象**、再用**公开数据与协议**问「换一把尺子故事会不会变」的手记；**格子里的**模拟与**磁盘上的**审计**分开写**，也不是「推翻量子力学」的檄文。若还有一处痒：从 `book/02-contents.md` 找章进正文，技术总图在 `docs/PROJECT_TECHNICAL_MONOGRAPH.md`；**不必会代码**也能抓住主张与边界。

50.1 开放问题与未完成的介质逻辑

本书刻意留下**未封口的缝**：

- **SI 映射**：许多旋钮仍是**无量纲**；介质参数与真实材料常数的**一一对应**未在书中完成。
- **浴与连续极限**：格子上「漏掉」的能量，**能否**、**在何条件下**与严格热力学变量挂钩——仍是**议程**。
- **GHZ / Bell**：档案层与 in-silico 层**并排**；**没有**宣称已在全参数空间**击败**标准量子叙事，只展示**尺子移动时故事如何变**。
- **涟漪量子 toy**：**v8** 与色散仍是**现象学闭合**；读者若看到「漂亮曲线」，请先问**门禁与失败记录**。
- **第十五章（规划）**：Bell 类数据上 **Alice/Bob 球面坐标联合拟合**——**文稿已占位，实现待作者有空再做**。

这些不是「以后随便说说」，而是**下一次迭代**的**入口坐标**。

全书之外的研究导航（仓库根）

- `docs/RESEARCH_MAP.md`：五层结构（直觉 →Bell/CHSH→ 工具 → 类比 → 定位），防失焦。
- `docs/PROJECT_TECHNICAL_MONOGRAPH.md`：与代码分层、实验树、解释边界对齐的「技术专论」。
- `unsolved_problems_collision/ROADMAP_7_PROBLEMS.md`：把「未解讨论」压成 **metrics + 对照 + 失败判据** 的示范。
- `docs/RIPPLE_QUANTUM_TESTS_INDEX.md`：涟漪量子 toy **唯一索引入口**。

本书正文文件索引：见 `book/02-contents.md`（含第 1–18 章、档案、后记与封面）。

50.2 交给后来者

我把**假大象**画到这里：**泡泡**是直觉，**格子**是脚手架，**公开数据**是考场，**协议**是监考规则。

若你接着画：

1. **先对齐定义**再对齐口号；
2. **模型内**与**审计**分开存档；
3. **失败**与**通过**同等入册。

书到这里可以合上一角，但**涟漪**还会继续扩散——碰到什么，产生什么新的涟漪，我**无法预知**，也不需要预知。

王辉•Hui Wang
2026

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51 Afterword

You just read a **fake elephant** built from **bubble intuition**, then pressed with **public data and protocols** —asking whether **changing the yardstick changes the story**. **On-grid** simulation and **on-disk** audit stay **separate**; this is not a tract to "overthrow quantum mechanics." If something still

itches: `book/02-contents.md` for chapter paths, `docs/PROJECT_TECHNICAL_MONOGRAPH.md` for the technical spine —**no coding required** to grasp the claims and boundaries.

51.1 Open questions and unfinished medium logic

The manuscript **leaves seams open on purpose**:

- **SI mapping**: many knobs are still **dimensionless**; a full **one-to-one** map from medium parameters to real material constants is **not finished** in-book.
- **Bath and continuum limits**: whether energy "lost" on the lattice **can**, and **under what conditions**, tie cleanly to strict thermodynamic variables —still **an agenda**.
- **GHZ / Bell**: **archive layer** and **in-silico layer** sit **side by side**; **no** claim to have **beaten** standard quantum narrative across all parameter space —only to show **how stories move when yardsticks move**.
- **Ripple-quantum toys**: **v8** and dispersion remain **phenomenological closure**; if a curve looks pretty, first ask **which gates it passed and where failures were logged**.
- **Chapter 15 (planned)**: **Joint Alice/Bob sphere fit on Bell-type data —manuscript placeholder; implementation when the author has bandwidth.**

These are **not** "mention later and forget"; they are **coordinates for the next iteration.**

Research navigation outside the manuscript (repo root)

- `docs/RESEARCH_MAP.md`: five-layer map (intuition → Bell/CHSH → tools → analogy → positioning) to avoid drift.
- `docs/PROJECT_TECHNICAL_MONOGRAPH.md`: "technical monograph" aligned with code layering, experiment tree, interpretive bounds.
- `unsolved_problems_collision/ROADMAP_7_PROBLEMS.md`: template that compresses open discussion into **metrics + controls + failure criteria**.
- `docs/RIPPLE_QUANTUM_TESTS_INDEX.md`: **single index entry** for ripple-quantum toys.

Manuscript file index: `book/02-contents.md` (Chapters 1–18, archives, afterword, cover).

51.2 To those who follow

I drew the **fake elephant** this far: **bubble** is intuition, **lattice** is scaffolding, **public data** is the exam hall, **protocol** is the invigilation rulebook.

If you keep drawing:

1. **Align definitions** before aligning slogans;
2. Archive **in-model** runs **separately** from **public-record audits**;
3. Log **failures** with the same care as **passes**.

The book can fold a corner here, but **ripples** keep spreading —what they touch, what new ripples they make, I **cannot foresee** and do not need to.

Hui Wang
2026

Afterword (English) —compact recap

The book deliberately leaves **open seams**: dimensionless knobs without a full SI map; lattice-to-continuum and heat-bath stories unfinished; Bell/GHZ split into **archive logic** vs **in-silico audits** — no claim to have closed all parameter space against standard quantum accounts, only to show **how narratives move when statistical objects move**. Chapter 15 is an **explicit placeholder** for a future **Bell-focused joint fit** of Alice/Bob **Poincaré-style coordinates** on public data; **not implemented yet**.

If you continue: **lock definitions before slogans**; archive **in-model** runs separately from **public-record audits**; log **failures** alongside **passes**.